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ЗАДАНИЕ 1

1) Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров

настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета

```
vIOS-L2-01>enable
```

```
vIOS-L2-01#show spanning-tree
```

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769

Address 0c34.c739.0000

Cost 4

Port 3 (GigabitEthernet0/2)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0cc8.d772.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Altn	BLK	4	128.1	Shr	
-------	------	-----	---	-------	-----	--

Gi0/1	Altn	BLK	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Root	FWD	4	128.3	Shr	
-------	------	-----	---	-------	-----	--

Gi0/3	Altn	BLK	4	128.4	Shr	
-------	------	-----	---	-------	-----	--

Gi1/0	Desg	FWD	4	128.5	Shr	
-------	------	-----	---	-------	-----	--

Gi1/1	Desg	FWD	4	128.6	Shr	
-------	------	-----	---	-------	-----	--

Gi1/2	Desg	FWD	4	128.7	Shr	
-------	------	-----	---	-------	-----	--

vIOS-L2-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

vIOS-L2-01(config)#spanning-tree vlan 1 priority 4096

vIOS-L2-01(config)#end

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cc8.d772.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 4097 (priority 4096 sys-id-ext 1)

Address 0cc8.d772.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr

Gi2/0 Desg FWD 4 128.9 Shr

vIOS-L2-01#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

Compressed configuration from 5277 bytes to 2026 bytes[OK]

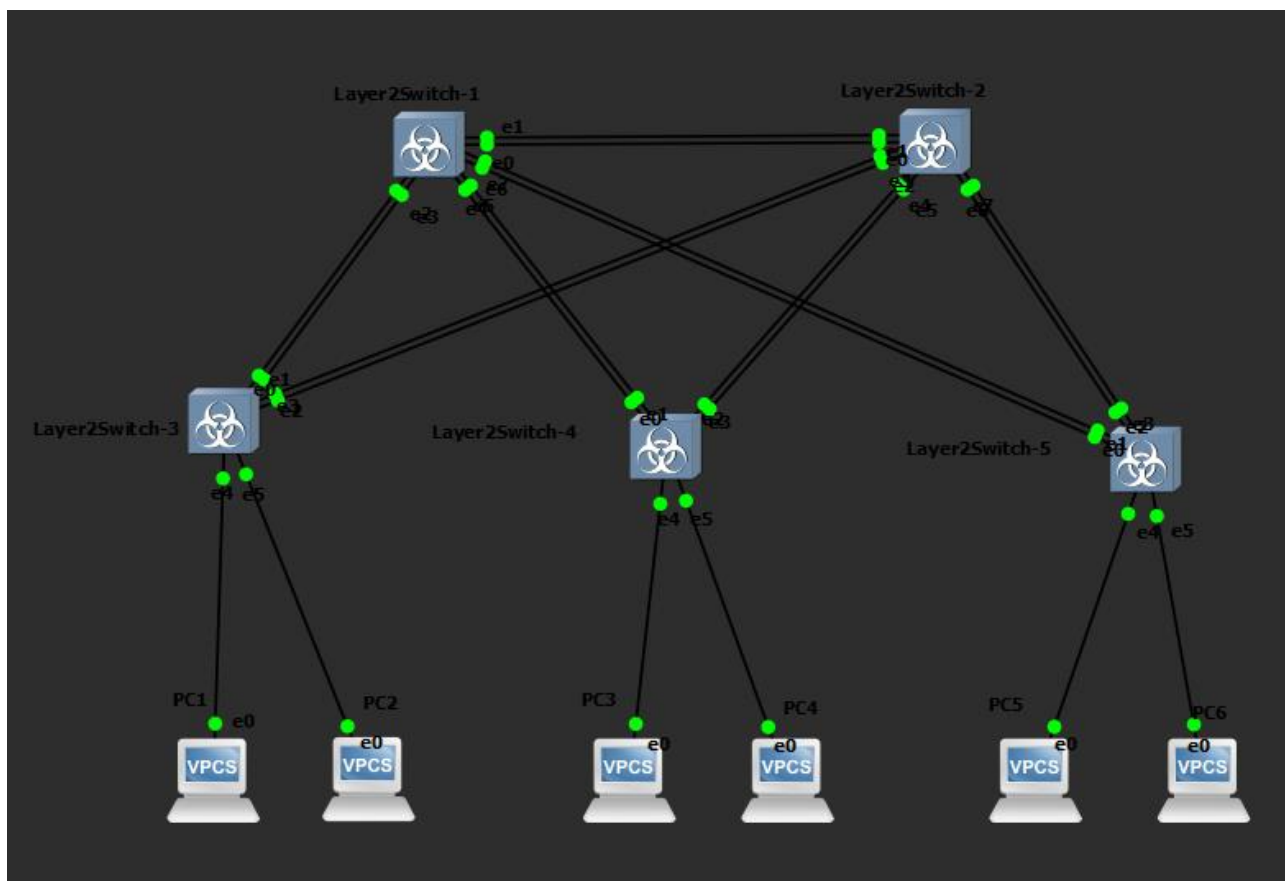


Рисунок 1 — Топология сети для исследования протокола STP

ЗАДАНИЕ 2

2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать

```
PC1> ip 192.168.10.1/24
```

```
Checking for duplicate address...
```

```
PC1 : 192.168.10.1 255.255.255.0
```

```
PC1> save
```

```
Saving startup configuration to startup.vpc
```

```
. done
```

```
PC1> show ip
```

```
NAME      : PC1[1]
```

```
IP/MASK    : 192.168.10.1/24
```

```
GATEWAY    : 0.0.0.0
```

```
DNS        :
```

```
MAC        : 00:50:79:66:68:00
```

```
LPORT      : 22662
```

```
RHOST:PORT : 127.0.0.1:22663
```

```
MTU        : 1500
```

```
PC2> ip 192.168.10.2/24
```

```
Checking for duplicate address...
```

```
PC2 : 192.168.10.2 255.255.255.0
```

```
PC2> save
```

```
Saving startup configuration to startup.vpc
```

. done

PC2> show ip

NAME : PC2[1]
IP/MASK : 192.168.10.2/24
GATEWAY : 0.0.0.0
DNS :
MAC : 00:50:79:66:68:01
LPORT : 22688
RHOST:PORT : 127.0.0.1:22689
MTU : 1500

PC3> ip 192.168.10.3/24

Checking for duplicate address...

PC3 : 192.168.10.3 255.255.255.0

PC3> save

Saving startup configuration to startup.vpc

. done

PC3> show ip

NAME : PC3[1]
IP/MASK : 192.168.10.3/24
GATEWAY : 0.0.0.0
DNS :
MAC : 00:50:79:66:68:02
LPORT : 22690

RHOST:PORT : 127.0.0.1:22691

MTU : 1500

PC4> ip 192.168.10.4/24

Checking for duplicate address...

PC4 : 192.168.10.4 255.255.255.0

PC4> save

Saving startup configuration to startup.vpc

. done

PC4> show ip

NAME : PC4[1]

IP/MASK : 192.168.10.4/24

GATEWAY : 0.0.0.0

DNS :

MAC : 00:50:79:66:68:03

LPORT : 22692

RHOST:PORT : 127.0.0.1:22693

MTU : 1500

PC5> ip 192.168.10.5/24

Checking for duplicate address...

PC5 : 192.168.10.5 255.255.255.0

PC5> save

Saving startup configuration to startup.vpc

. done

PC5> show ip

NAME : PC5[1]
IP/MASK : 192.168.10.5/24
GATEWAY : 0.0.0.0
DNS :
MAC : 00:50:79:66:68:04
LPORT : 22694
RHOST:PORT : 127.0.0.1:22695
MTU : 1500

PC6> ip 192.168.10.6/24

Checking for duplicate address...

PC6 : 192.168.10.6 255.255.255.0

PC6> save

Saving startup configuration to startup.vpc

. done

PC6> show ip

NAME : PC6[1]
IP/MASK : 192.168.10.6/24
GATEWAY : 0.0.0.0
DNS :
MAC : 00:50:79:66:68:05
LPORT : 22696

RHOST:PORT : 127.0.0.1:22697

MTU : 1500

PC1> ping 192.168.10.2

84 bytes from 192.168.10.2 icmp_seq=1 ttl=64 time=3.012 ms

84 bytes from 192.168.10.2 icmp_seq=2 ttl=64 time=6.846 ms

84 bytes from 192.168.10.2 icmp_seq=3 ttl=64 time=5.362 ms

84 bytes from 192.168.10.2 icmp_seq=4 ttl=64 time=9.610 ms

84 bytes from 192.168.10.2 icmp_seq=5 ttl=64 time=4.240 ms

PC1> ping 192.168.10.3

84 bytes from 192.168.10.3 icmp_seq=1 ttl=64 time=18.507 ms

84 bytes from 192.168.10.3 icmp_seq=2 ttl=64 time=8.556 ms

84 bytes from 192.168.10.3 icmp_seq=3 ttl=64 time=10.837 ms

84 bytes from 192.168.10.3 icmp_seq=4 ttl=64 time=27.353 ms

84 bytes from 192.168.10.3 icmp_seq=5 ttl=64 time=7.680 ms

PC1> ping 192.168.10.4

84 bytes from 192.168.10.4 icmp_seq=1 ttl=64 time=25.263 ms

84 bytes from 192.168.10.4 icmp_seq=2 ttl=64 time=15.302 ms

84 bytes from 192.168.10.4 icmp_seq=3 ttl=64 time=24.843 ms

84 bytes from 192.168.10.4 icmp_seq=4 ttl=64 time=18.314 ms

84 bytes from 192.168.10.4 icmp_seq=5 ttl=64 time=10.408 ms

PC1> ping 192.168.10.5

84 bytes from 192.168.10.5 icmp_seq=1 ttl=64 time=15.769 ms
84 bytes from 192.168.10.5 icmp_seq=2 ttl=64 time=13.129 ms
84 bytes from 192.168.10.5 icmp_seq=3 ttl=64 time=6.645 ms
84 bytes from 192.168.10.5 icmp_seq=4 ttl=64 time=10.288 ms
84 bytes from 192.168.10.5 icmp_seq=5 ttl=64 time=25.624 ms

PC1> ping 192.168.10.6

84 bytes from 192.168.10.6 icmp_seq=1 ttl=64 time=8.118 ms
84 bytes from 192.168.10.6 icmp_seq=2 ttl=64 time=9.811 ms
84 bytes from 192.168.10.6 icmp_seq=3 ttl=64 time=10.707 ms
84 bytes from 192.168.10.6 icmp_seq=4 ttl=64 time=11.894 ms
84 bytes from 192.168.10.6 icmp_seq=5 ttl=64 time=5.679 ms

PC2> ping 192.168.10.3

84 bytes from 192.168.10.3 icmp_seq=1 ttl=64 time=8.461 ms
84 bytes from 192.168.10.3 icmp_seq=2 ttl=64 time=13.251 ms
84 bytes from 192.168.10.3 icmp_seq=3 ttl=64 time=13.477 ms
84 bytes from 192.168.10.3 icmp_seq=4 ttl=64 time=13.448 ms
84 bytes from 192.168.10.3 icmp_seq=5 ttl=64 time=11.522 ms

PC2> ping 192.168.10.4

84 bytes from 192.168.10.4 icmp_seq=1 ttl=64 time=15.470 ms
84 bytes from 192.168.10.4 icmp_seq=2 ttl=64 time=8.540 ms
84 bytes from 192.168.10.4 icmp_seq=3 ttl=64 time=11.932 ms
84 bytes from 192.168.10.4 icmp_seq=4 ttl=64 time=19.985 ms
84 bytes from 192.168.10.4 icmp_seq=5 ttl=64 time=10.065 ms

PC2> ping 192.168.10.5

84 bytes from 192.168.10.5 icmp_seq=1 ttl=64 time=9.451 ms
84 bytes from 192.168.10.5 icmp_seq=2 ttl=64 time=5.749 ms
84 bytes from 192.168.10.5 icmp_seq=3 ttl=64 time=5.881 ms
84 bytes from 192.168.10.5 icmp_seq=4 ttl=64 time=19.639 ms
84 bytes from 192.168.10.5 icmp_seq=5 ttl=64 time=11.156 ms

PC2> ping 192.168.10.6

84 bytes from 192.168.10.6 icmp_seq=1 ttl=64 time=19.132 ms
84 bytes from 192.168.10.6 icmp_seq=2 ttl=64 time=4.188 ms
84 bytes from 192.168.10.6 icmp_seq=3 ttl=64 time=13.543 ms
84 bytes from 192.168.10.6 icmp_seq=4 ttl=64 time=13.455 ms
84 bytes from 192.168.10.6 icmp_seq=5 ttl=64 time=21.767 ms

PC3> ping 192.168.10.4

84 bytes from 192.168.10.4 icmp_seq=1 ttl=64 time=4.762 ms
84 bytes from 192.168.10.4 icmp_seq=2 ttl=64 time=1.633 ms
84 bytes from 192.168.10.4 icmp_seq=3 ttl=64 time=6.888 ms
84 bytes from 192.168.10.4 icmp_seq=4 ttl=64 time=0.918 ms
84 bytes from 192.168.10.4 icmp_seq=5 ttl=64 time=1.662 ms

PC3> ping 192.168.10.5

84 bytes from 192.168.10.5 icmp_seq=1 ttl=64 time=14.628 ms
84 bytes from 192.168.10.5 icmp_seq=2 ttl=64 time=13.074 ms

84 bytes from 192.168.10.5 icmp_seq=3 ttl=64 time=16.995 ms
84 bytes from 192.168.10.5 icmp_seq=4 ttl=64 time=16.402 ms
84 bytes from 192.168.10.5 icmp_seq=5 ttl=64 time=18.525 ms

PC3> ping 192.168.10.6

84 bytes from 192.168.10.6 icmp_seq=1 ttl=64 time=11.949 ms
84 bytes from 192.168.10.6 icmp_seq=2 ttl=64 time=17.984 ms
84 bytes from 192.168.10.6 icmp_seq=3 ttl=64 time=11.390 ms
84 bytes from 192.168.10.6 icmp_seq=4 ttl=64 time=29.447 ms
84 bytes from 192.168.10.6 icmp_seq=5 ttl=64 time=22.980 ms

PC4> ping 192.168.10.5

84 bytes from 192.168.10.5 icmp_seq=1 ttl=64 time=9.322 ms
84 bytes from 192.168.10.5 icmp_seq=2 ttl=64 time=9.469 ms
84 bytes from 192.168.10.5 icmp_seq=3 ttl=64 time=28.402 ms
84 bytes from 192.168.10.5 icmp_seq=4 ttl=64 time=11.168 ms
84 bytes from 192.168.10.5 icmp_seq=5 ttl=64 time=20.497 ms

PC4> ping 192.168.10.6

84 bytes from 192.168.10.6 icmp_seq=1 ttl=64 time=17.119 ms
84 bytes from 192.168.10.6 icmp_seq=2 ttl=64 time=10.867 ms
84 bytes from 192.168.10.6 icmp_seq=3 ttl=64 time=12.207 ms
84 bytes from 192.168.10.6 icmp_seq=4 ttl=64 time=12.139 ms
84 bytes from 192.168.10.6 icmp_seq=5 ttl=64 time=5.667 ms

```
PC5> ping 192.168.10.6
```

```
84 bytes from 192.168.10.6 icmp_seq=1 ttl=64 time=8.502 ms
```

```
84 bytes from 192.168.10.6 icmp_seq=2 ttl=64 time=2.952 ms
```

```
84 bytes from 192.168.10.6 icmp_seq=3 ttl=64 time=3.664 ms
```

```
84 bytes from 192.168.10.6 icmp_seq=4 ttl=64 time=3.641 ms
```

```
84 bytes from 192.168.10.6 icmp_seq=5 ttl=64 time=7.766 ms
```

ЗАДАНИЕ 3

3) На изображении схемы отметить VID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл

```
vIOS-L2-01>show spanning-tree vlan 1
```

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cc8.d772.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 4097 (priority 4096 sys-id-ext 1)

Address 0cc8.d772.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi2/0	Desg FWD 4	128.9	Shr
-------	------------	-------	-----

vIOS-L2-01>enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

 Address 0cc8.d772.0000

 Cost 4

 Port 1 (GigabitEthernet0/0)

 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

 Address 0cab.92c5.0000

 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

 Aging Time 300 sec

Interface	Role Sts Cost	Prio.Nbr	Type
-----------	---------------	----------	------

Gi0/0	Root FWD 4	128.1	Shr
-------	------------	-------	-----

Gi0/1	Altn BLK 4	128.2	Shr
-------	------------	-------	-----

Gi0/2	Altn BLK 4	128.3	Shr
-------	------------	-------	-----

Gi0/3	Altn BLK 4	128.4	Shr
-------	------------	-------	-----

Gi1/0	Desg FWD 4	128.5	Shr
-------	------------	-------	-----

Gi1/1	Desg FWD 4	128.6	Shr
-------	------------	-------	-----

Gi1/2	Altn BLK 4	128.7	Shr
-------	------------	-------	-----

Gi1/3	Altn BLK 4	128.8	Shr
Gi2/0	Desg FWD 4	128.9	Shr

vIOS-L2-01>enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cc8.d772.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c34.c739.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role Sts Cost	Prio.Nbr	Type
-----------	---------------	----------	------

Gi0/0	Root FWD 4	128.1	Shr
Gi0/1	Altn BLK 4	128.2	Shr
Gi0/2	Desg FWD 4	128.3	Shr
Gi0/3	Desg FWD 4	128.4	Shr
Gi1/0	Desg FWD 4	128.5	Shr
Gi1/1	Desg FWD 4	128.6	Shr

vIOS-L2-01#enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cc8.d772.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0cd8.c8fe.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Root	FWD	4	128.1	Shr	
-------	------	-----	---	-------	-----	--

Gi0/1	Altn	BLK	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Altn	BLK	4	128.3	Shr	
-------	------	-----	---	-------	-----	--

Gi0/3	Altn	BLK	4	128.4	Shr	
-------	------	-----	---	-------	-----	--

Gi1/0	Desg	FWD	4	128.5	Shr	
-------	------	-----	---	-------	-----	--

Gi1/1	Desg	FWD	4	128.6	Shr	
-------	------	-----	---	-------	-----	--

vIOS-L2-01>enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cc8.d772.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c43.0bfe.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Root	FWD	4	128.1	Shr	
-------	------	-----	---	-------	-----	--

Gi0/1	Altn	BLK	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Desg	FWD	4	128.3	Shr	
-------	------	-----	---	-------	-----	--

Gi0/3	Desg	FWD	4	128.4	Shr	
-------	------	-----	---	-------	-----	--

Gi1/0	Desg	FWD	4	128.5	Shr	
-------	------	-----	---	-------	-----	--

Gi1/1	Desg	FWD	4	128.6	Shr	
-------	------	-----	---	-------	-----	--

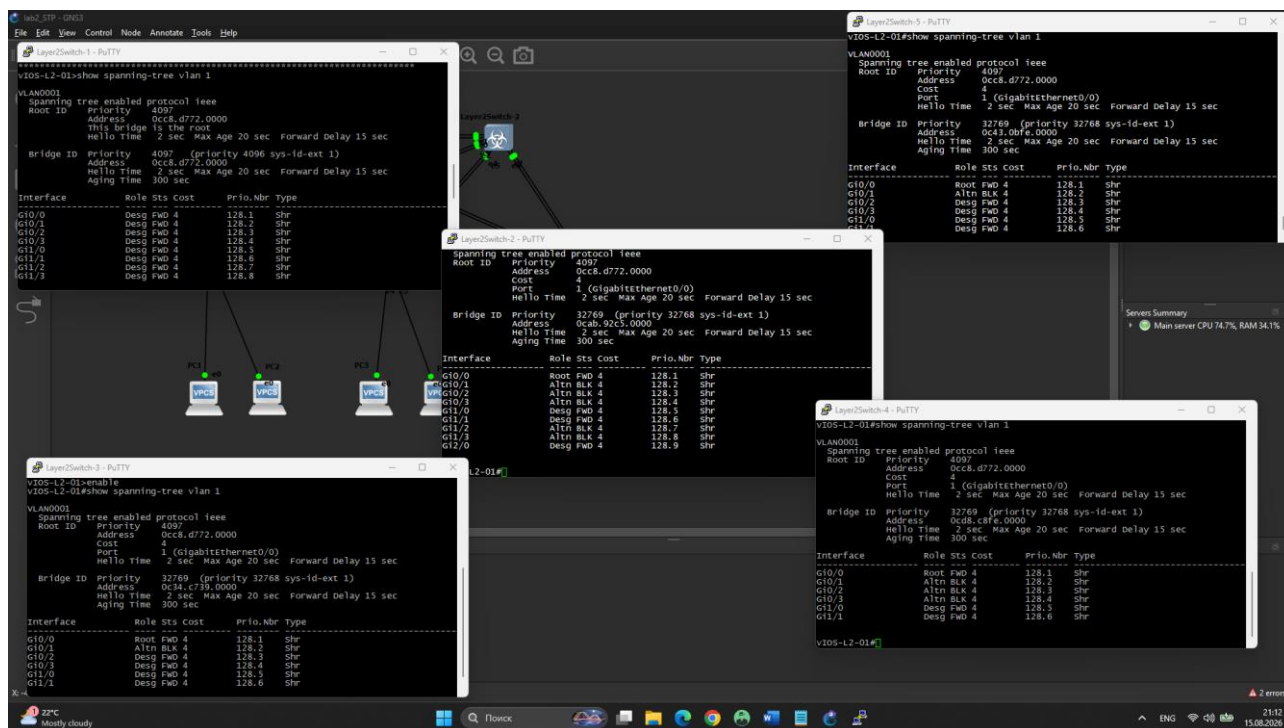


Рисунок 2 — Результаты проверки состояния STP на коммутаторах сети

vIOS-L2-01>show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
vIOS-L2-01	Gig 0/0	160	R S I	IOSv	Gig 0/0
vIOS-L2-01	Gig 0/1	164	R S I	IOSv	Gig 0/1
vIOS-L2-01	Gig 0/2	170	R S I	IOSv	Gig 0/0
vIOS-L2-01	Gig 0/3	147	R S I	IOSv	Gig 0/1
vIOS-L2-01	Gig 1/0	164	R S I	IOSv	Gig 0/0
vIOS-L2-01	Gig 1/1	178	R S I	IOSv	Gig 0/1
vIOS-L2-01	Gig 1/2	157	R S I	IOSv	Gig 0/0
vIOS-L2-01	Gig 1/3	158	R S I	IOSv	Gig 0/1

Total cdp entries displayed : 8

vIOS-L2-01>show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
vIOS-L2-01	Gig 0/0	148	R S I IOSv		Gig 0/0
vIOS-L2-01	Gig 0/3	135	R S I IOSv		Gig 0/3
vIOS-L2-01	Gig 0/2	153	R S I IOSv		Gig 0/2
vIOS-L2-01	Gig 0/1	137	R S I IOSv		Gig 0/1
vIOS-L2-01	Gig 1/0	143	R S I IOSv		Gig 0/2
vIOS-L2-01	Gig 1/1	137	R S I IOSv		Gig 0/3
vIOS-L2-01	Gig 1/2	135	R S I IOSv		Gig 0/2
vIOS-L2-01	Gig 1/3	146	R S I IOSv		Gig 0/3

Total cdp entries displayed : 8

vIOS-L2-01>show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
vIOS-L2-01	Gig 0/2	166	R S I IOSv		Gig 0/2
vIOS-L2-01	Gig 0/3	118	R S I IOSv		Gig 0/3

vIOS-L2-01	Gig 0/1	153	R S I IOSv	Gig 0/3
vIOS-L2-01	Gig 0/0	120	R S I IOSv	Gig 0/2

Total cdp entries displayed : 4

vIOS-L2-01#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
 S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
 D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
vIOS-L2-01	Gig 0/2	143	R S I IOSv		Gig 1/0
vIOS-L2-01	Gig 0/3	137	R S I IOSv		Gig 1/1
vIOS-L2-01	Gig 0/1	125	R S I IOSv		Gig 1/1
vIOS-L2-01	Gig 0/0	142	R S I IOSv		Gig 1/0

Total cdp entries displayed : 4

vIOS-L2-01#show cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
 S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
 D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
vIOS-L2-01	Gig 0/2	141	R S I IOSv		Gig 1/2
vIOS-L2-01	Gig 0/3	179	R S I IOSv		Gig 1/3
vIOS-L2-01	Gig 0/1	175	R S I IOSv		Gig 1/3
vIOS-L2-01	Gig 0/0	133	R S I IOSv		Gig 1/2

Total cdp entries displayed : 4

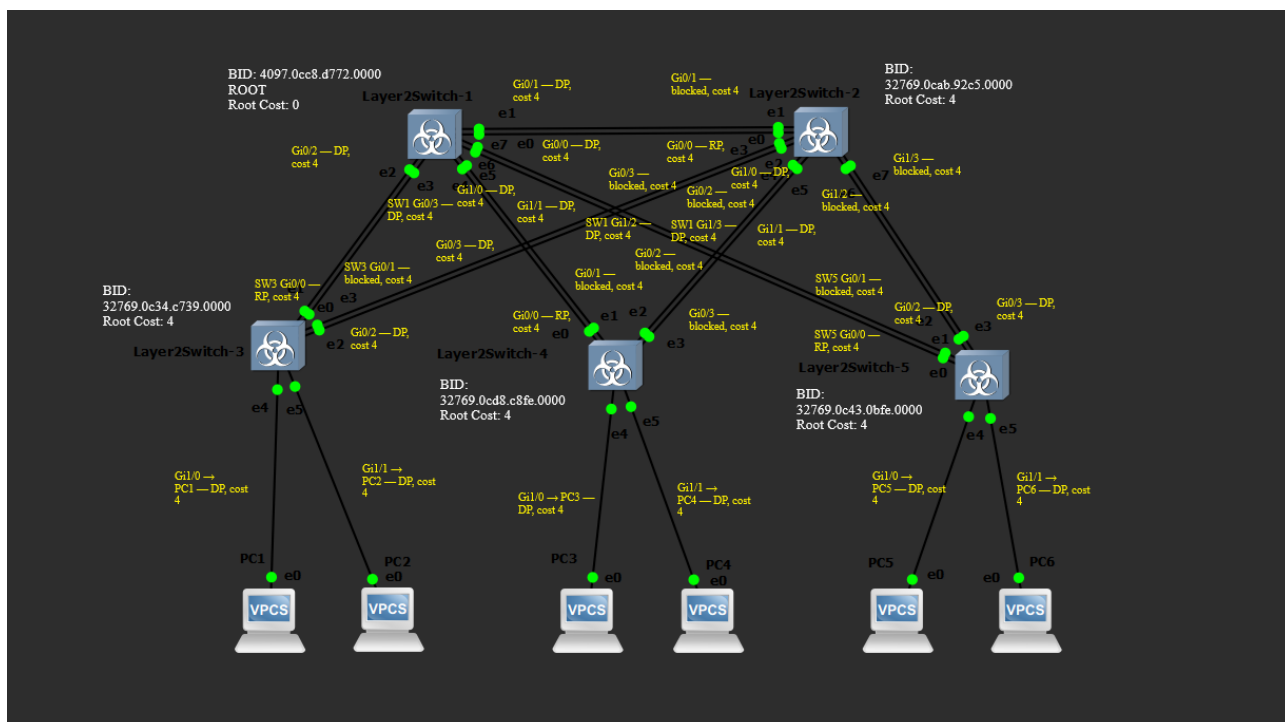


Рисунок 3 — Топология STP с BID коммутаторов, ролями портов и стоимостями маршрутов

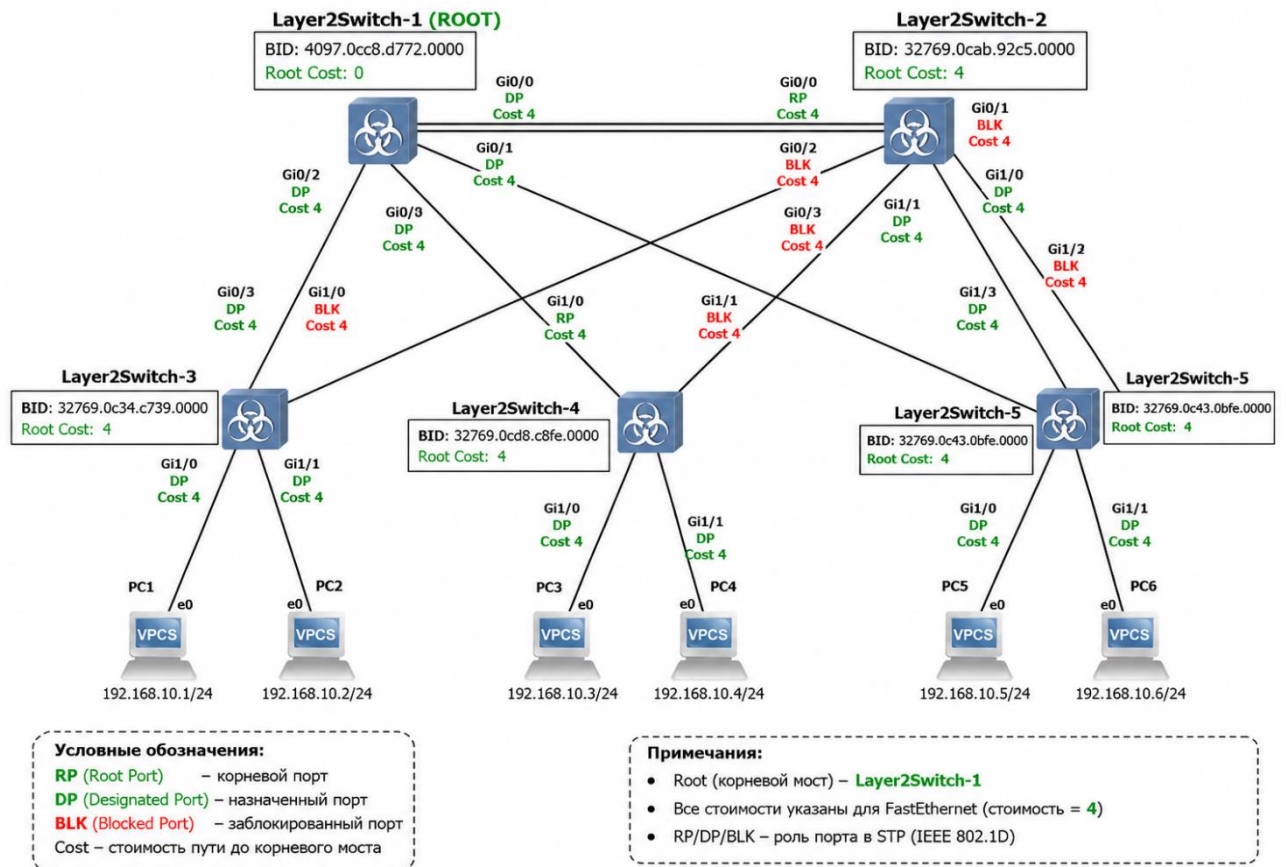


Рисунок 3.1 — Топология STP с BID коммутаторов, ролями портов и стоимостями маршрутов

ЗАДАНИЕ 4

4) При помощи Wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет

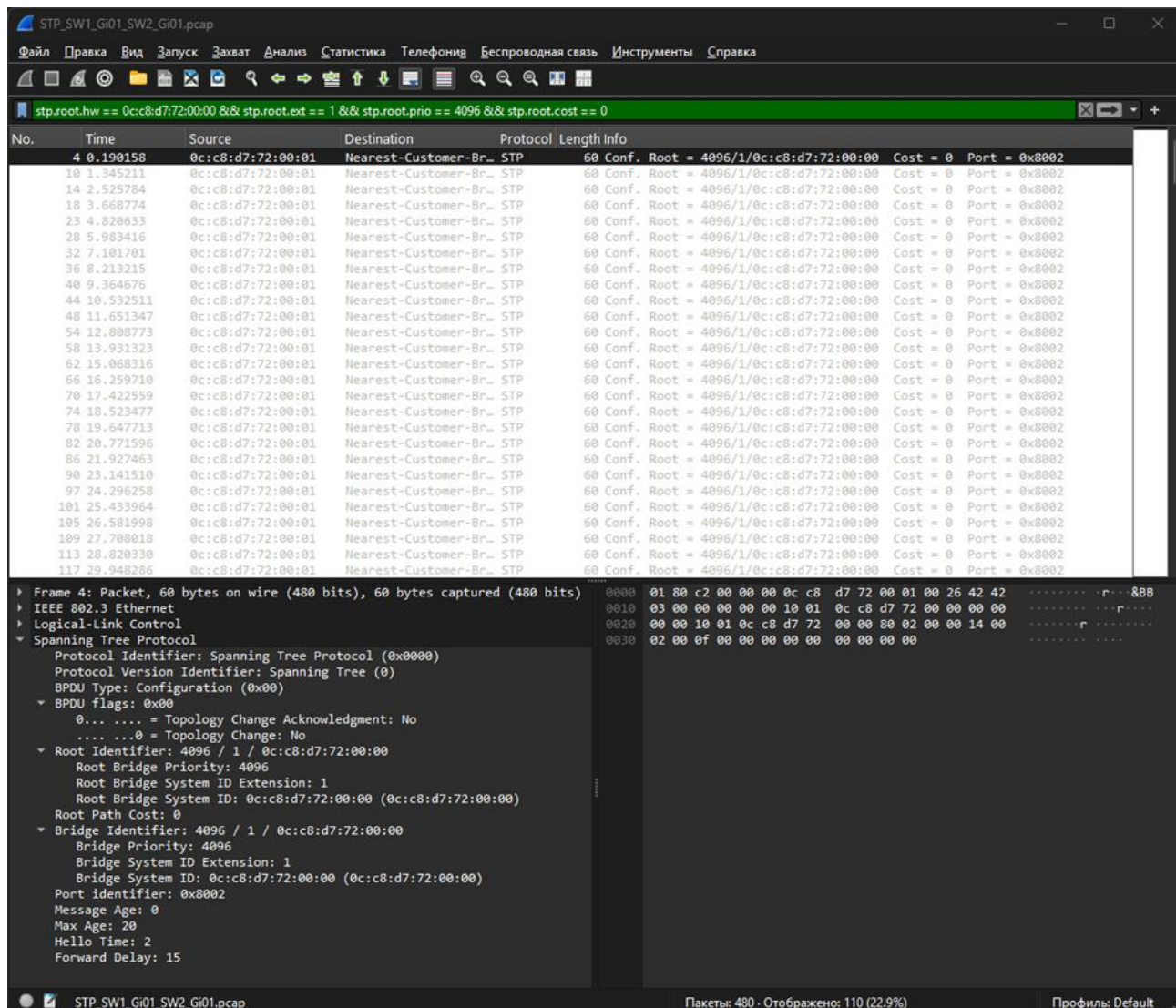


Рисунок 4 — STP Hello BPDU на линке Layer2Switch-1 Gi0/1 — Layer2Switch-2 Gi0/1

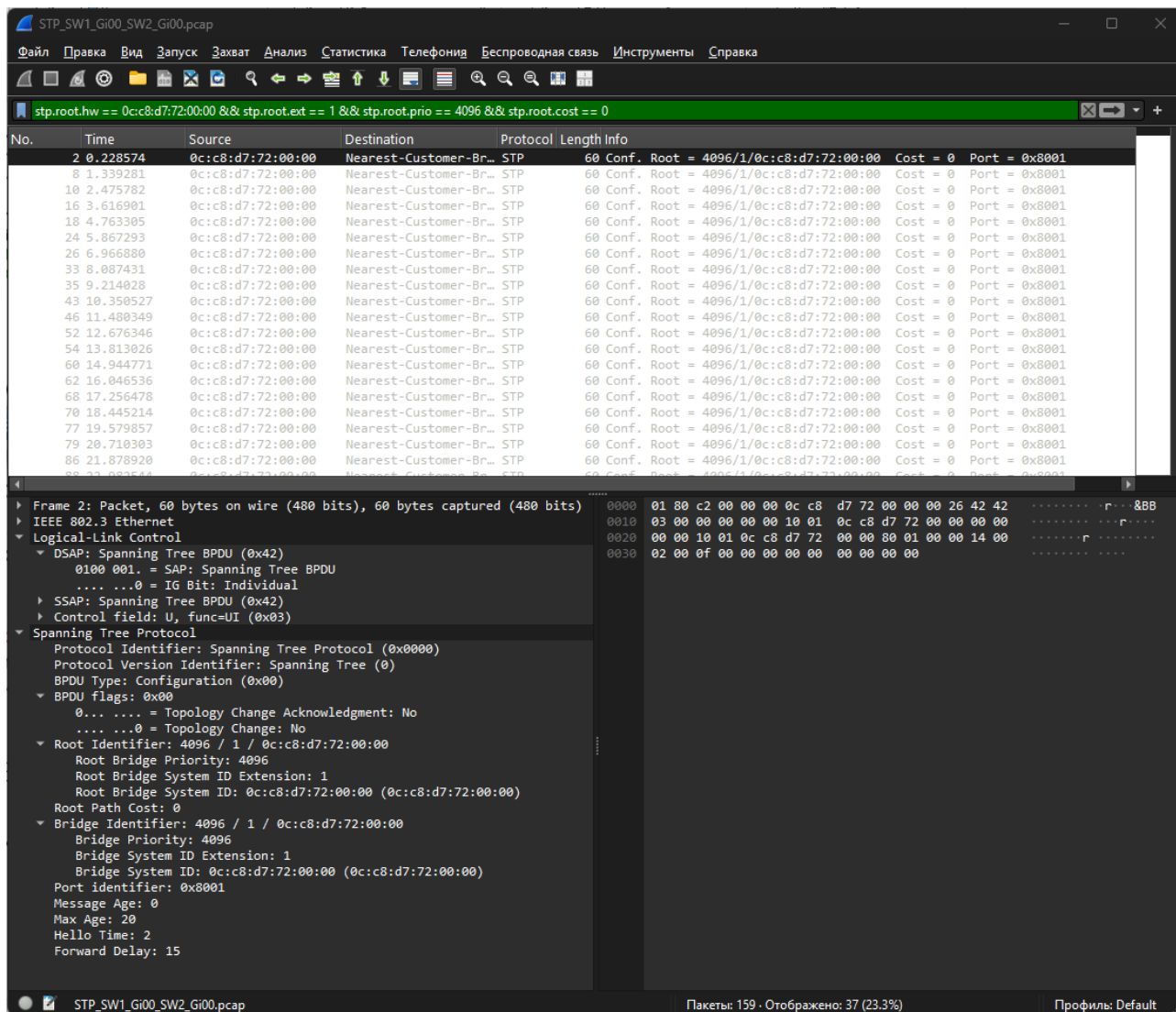


Рисунок 5 — STP Hello BPDU на линке Layer2Switch-1 Gi0/0 — Layer2Switch-2 Gi0/0

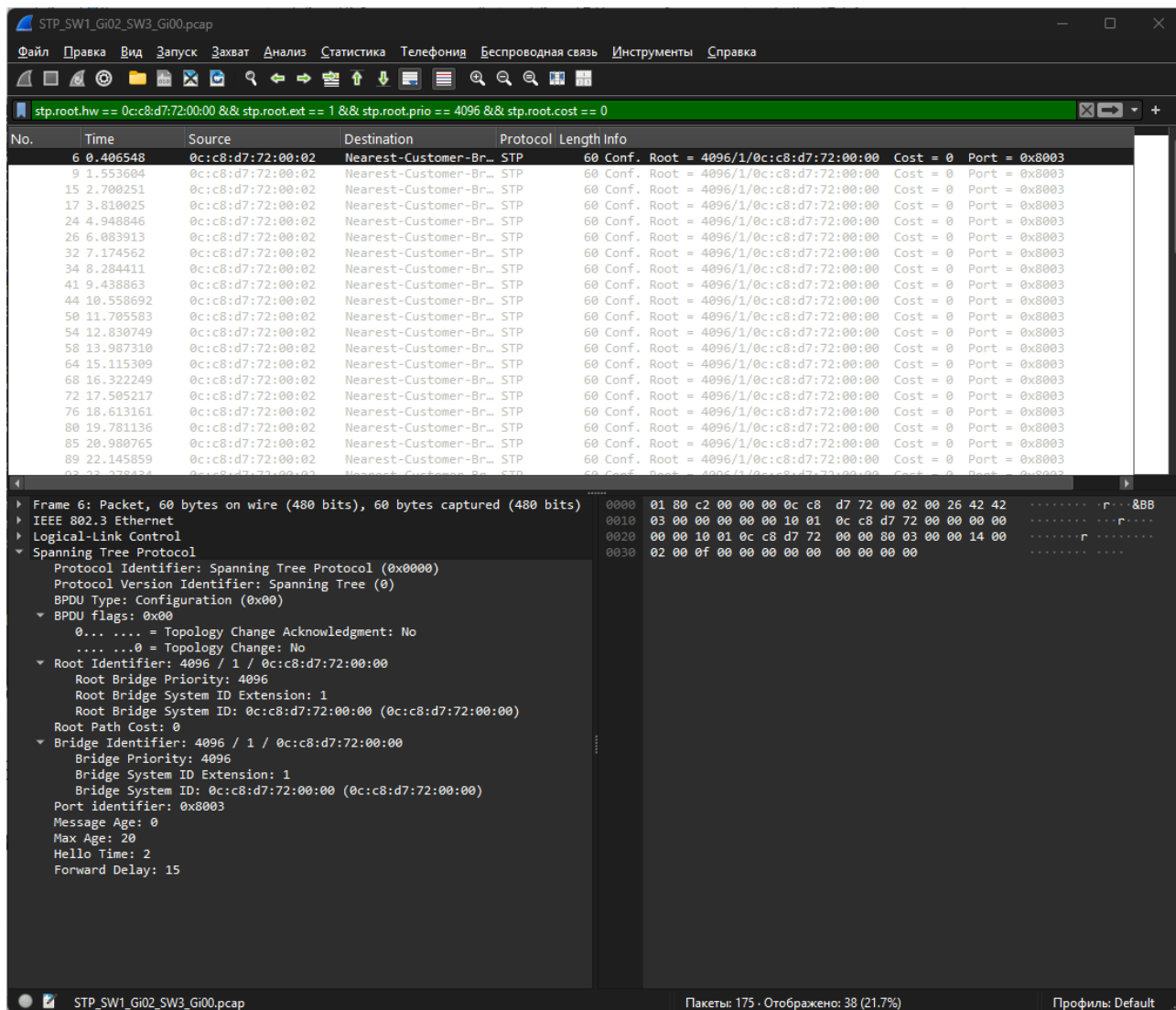


Рисунок 6 — STP Hello BPDU на линке Layer2Switch-1 Gi0/2 — Layer2Switch-3 Gi0/0

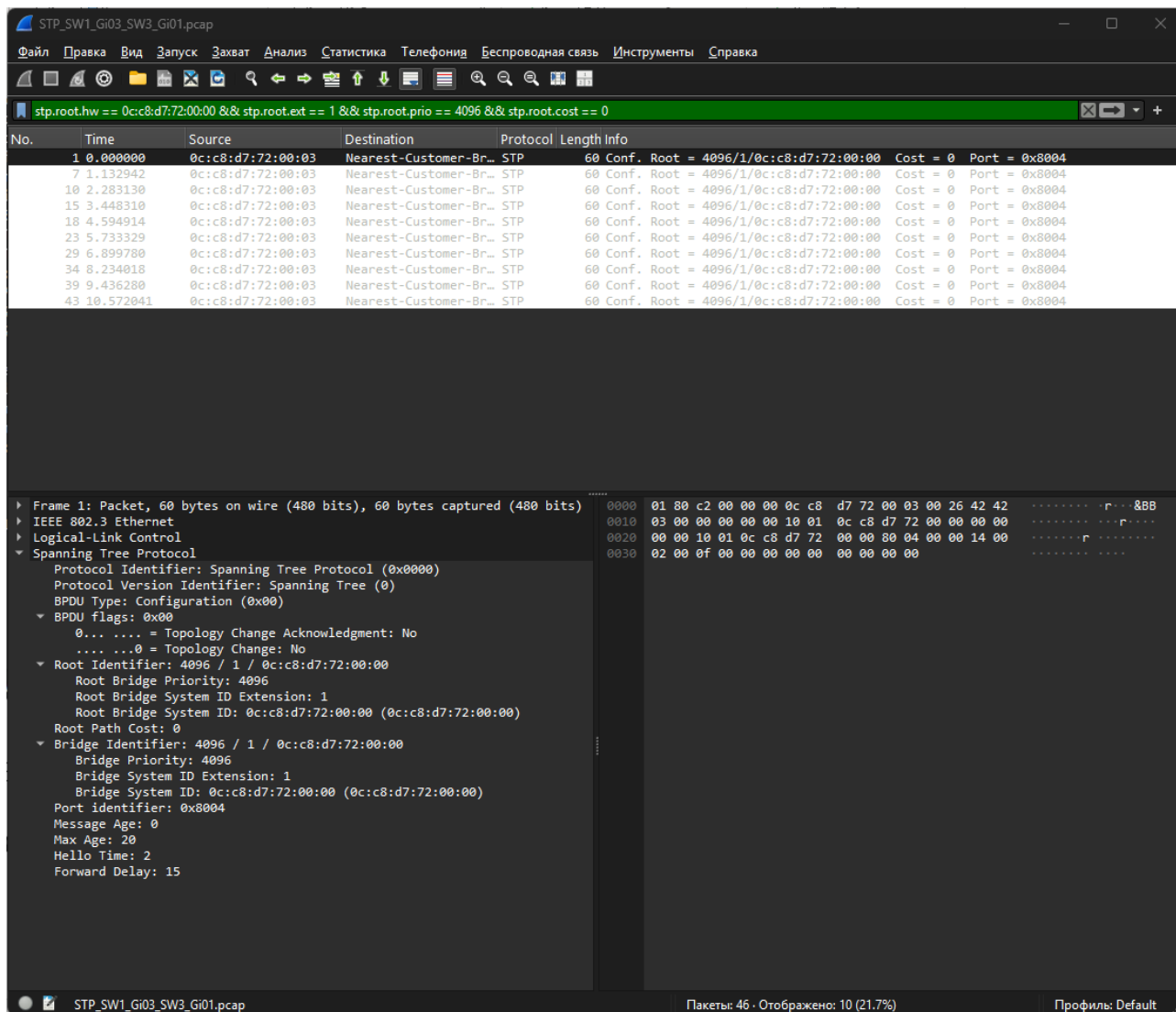


Рисунок 7 — STP Hello BPDU на линке Layer2Switch-1 Gi0/3 — Layer2Switch-3 Gi0/1

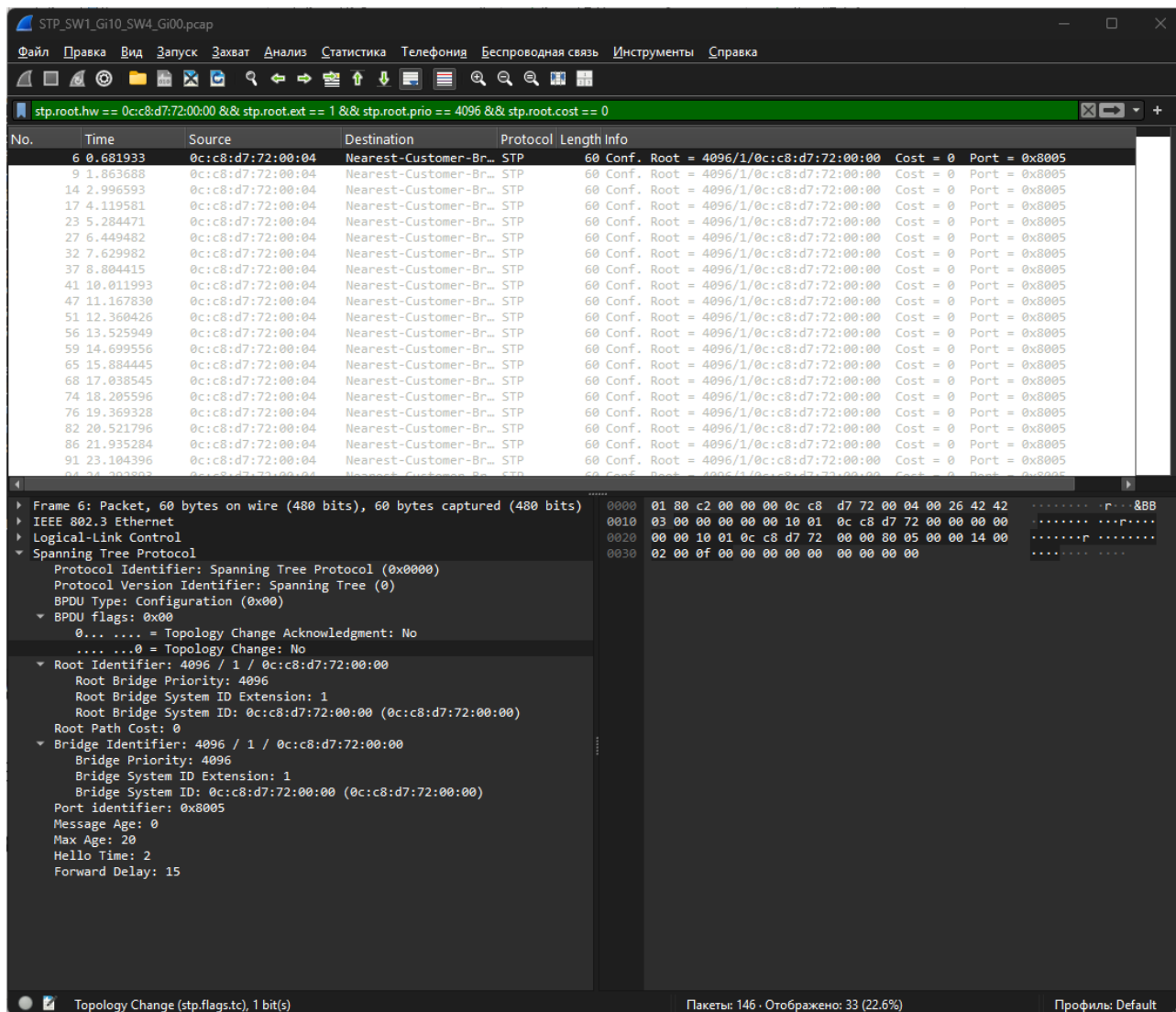


Рисунок 8 — STP Hello BPDU на линке Layer2Switch-1 Gi1/0 — Layer2Switch-4 Gi0/0

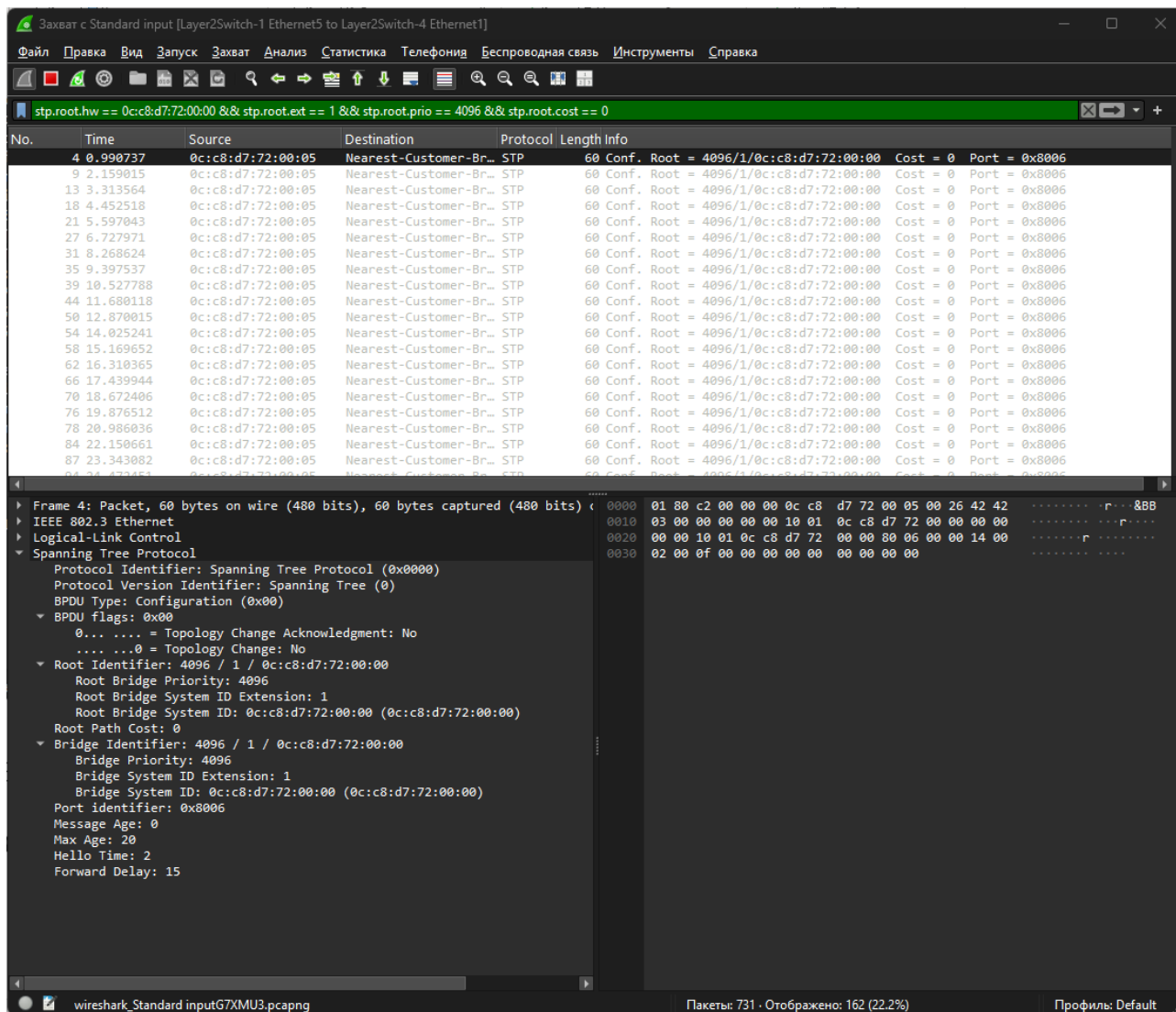


Рисунок 9 — STP Hello BPDU на линке Layer2Switch-1 Gi1/1 — Layer2Switch-4 Gi0/1

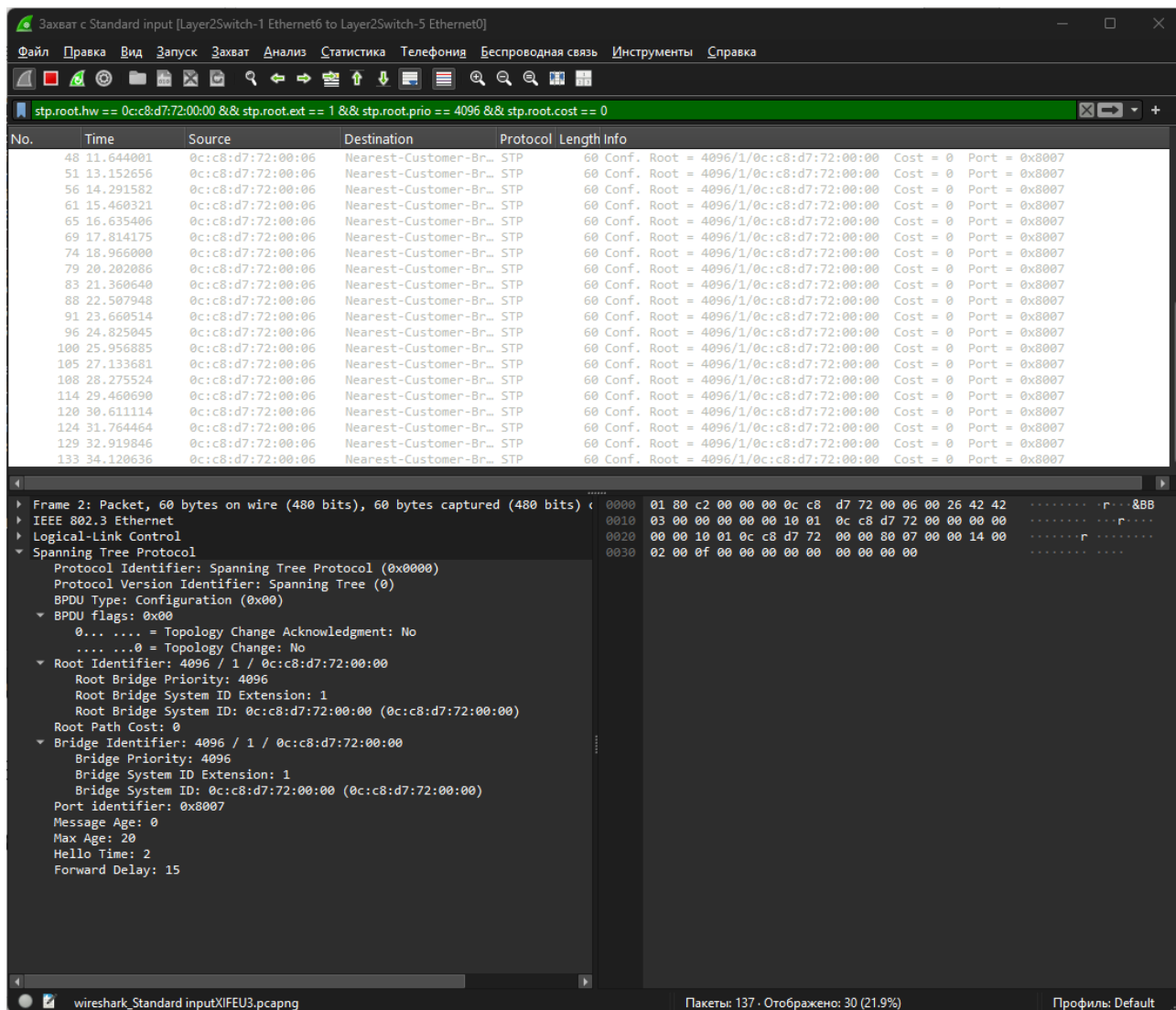


Рисунок 10 — STP Hello BPD на линке Layer2Switch-1 Gi1/2 — Layer2Switch-5 Gi0/0

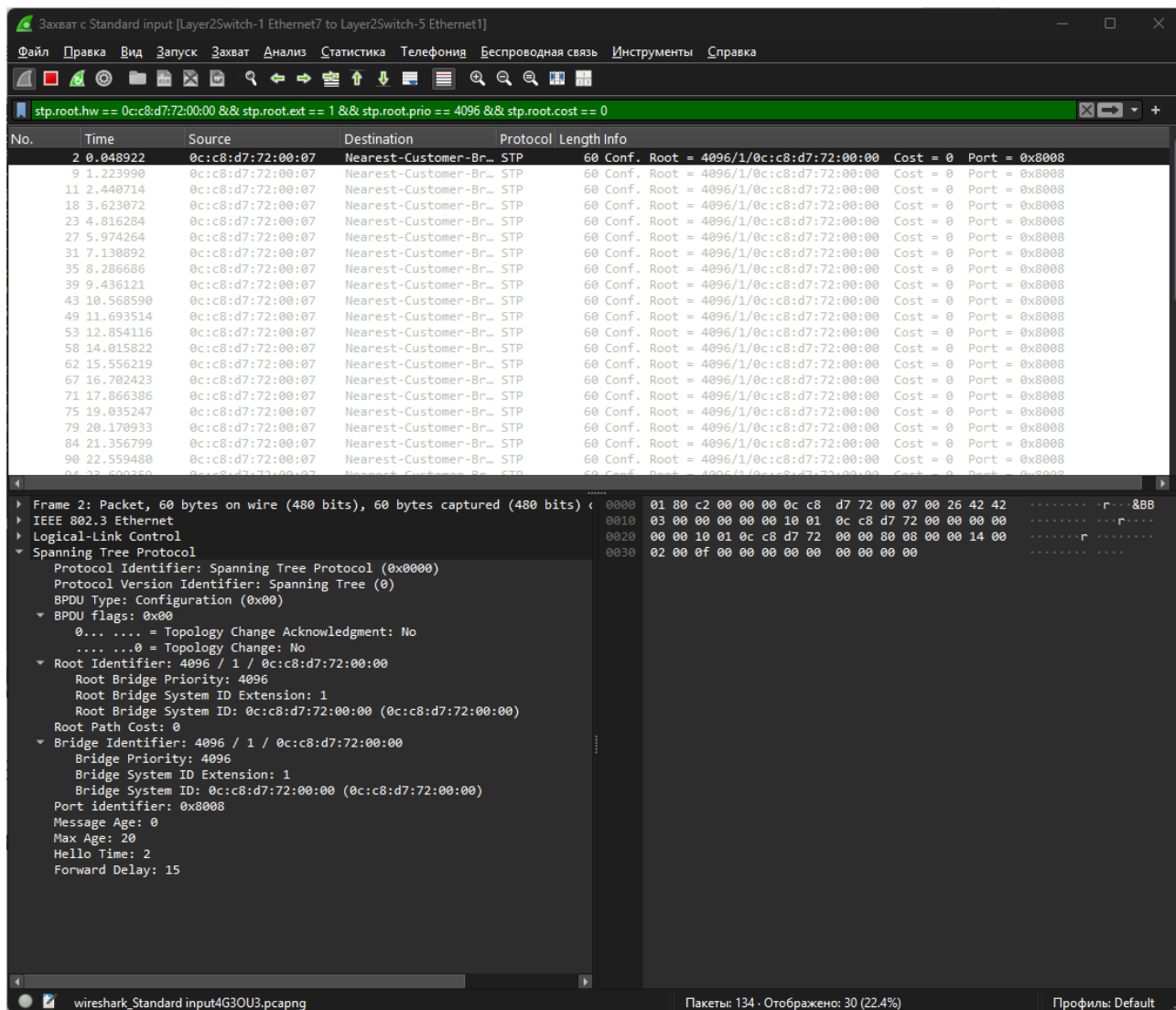


Рисунок 11 — STP Hello BPD на линке Layer2Switch-1 Gi1/3 — Layer2Switch-5 Gi0/1

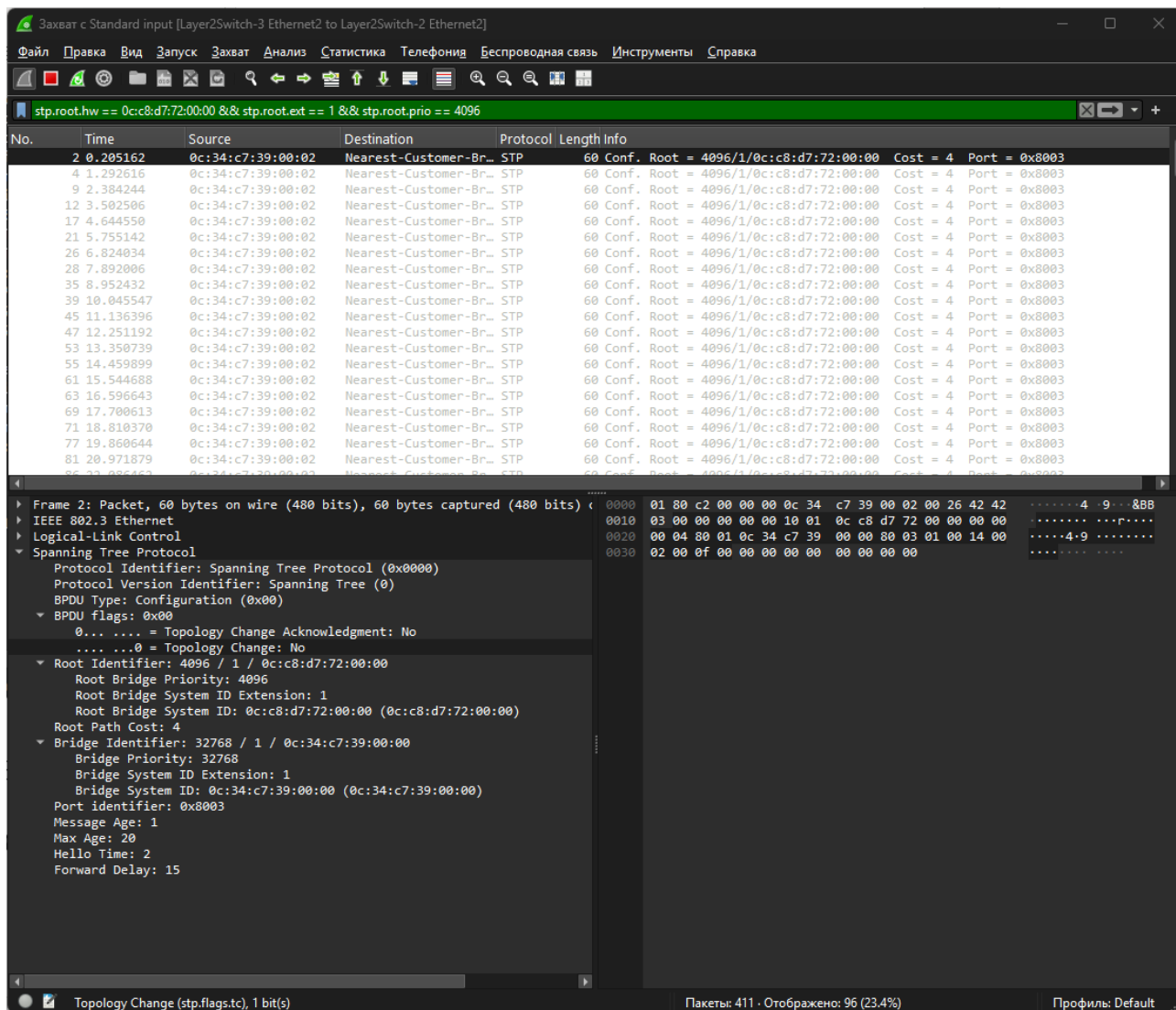


Рисунок 12 — STP Hello BPDU на линке Layer2Switch-2 Gi0/2 — Layer2Switch-3 Gi0/2

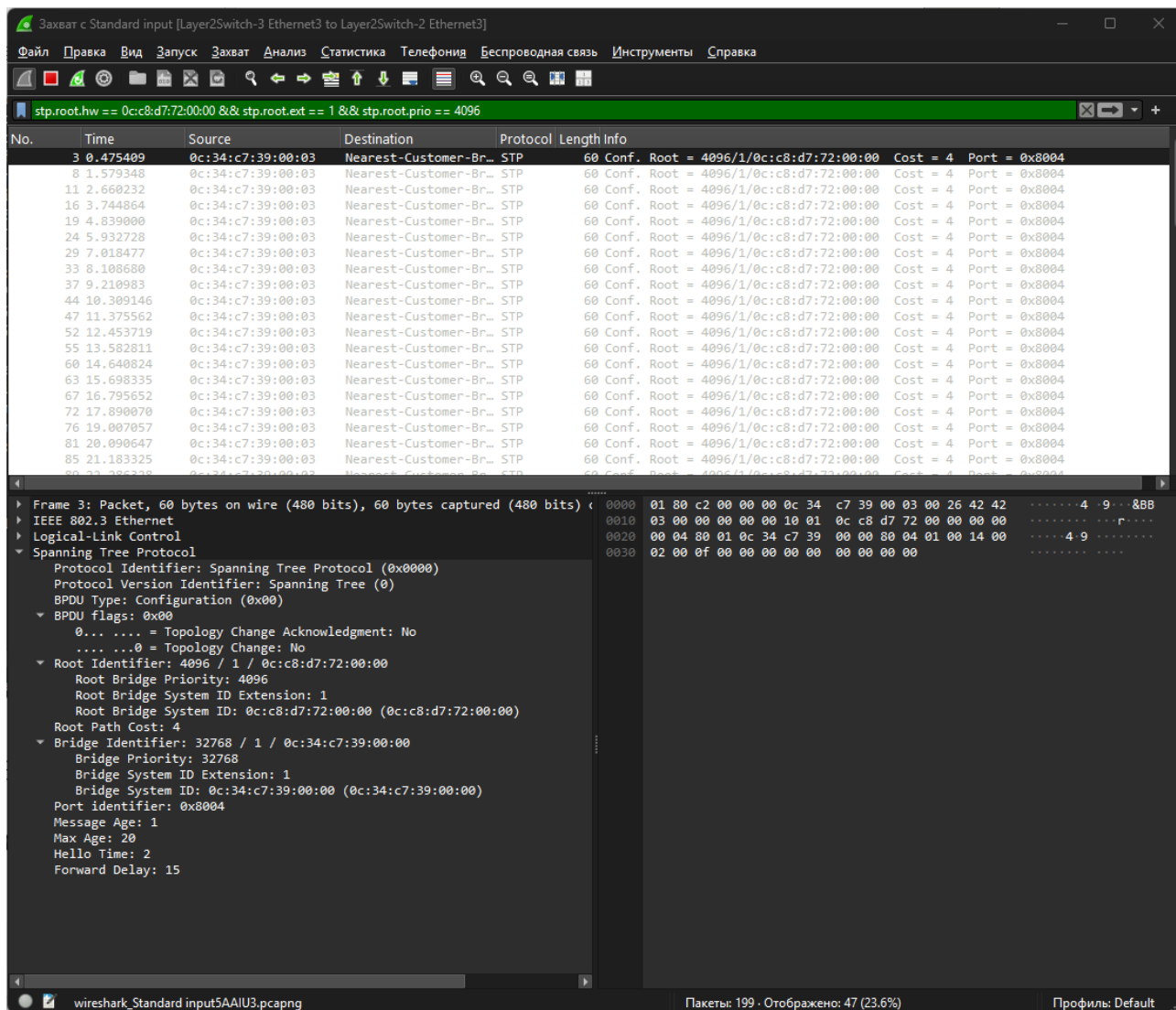


Рисунок 13 — STP Hello BPDU на линке Layer2Switch-2 Gi0/3 — Layer2Switch-3 Gi0/3

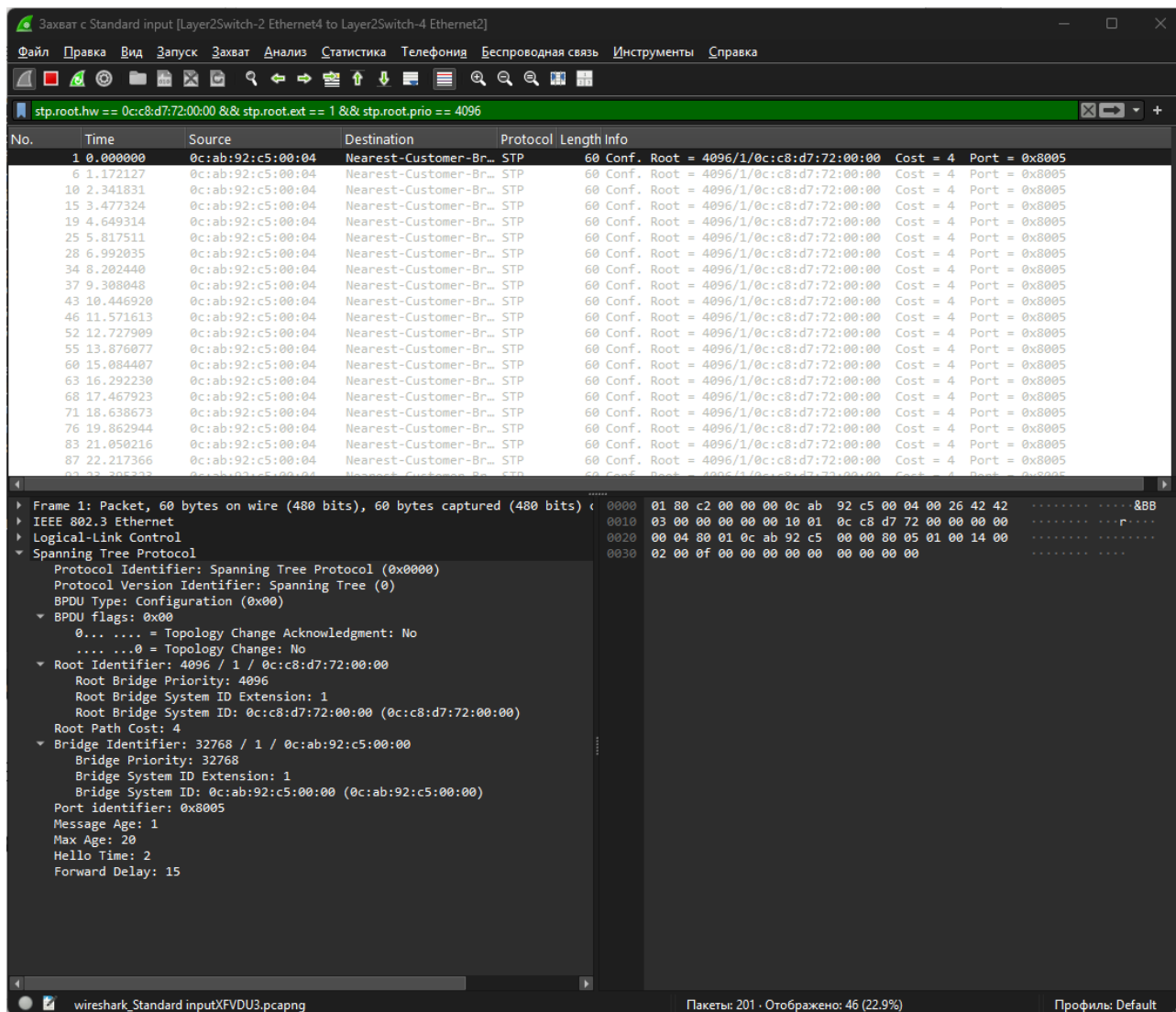


Рисунок 14 — STP Hello BPDU на линке Layer2Switch-2 Gi1/0 — Layer2Switch-4 Gi0/2

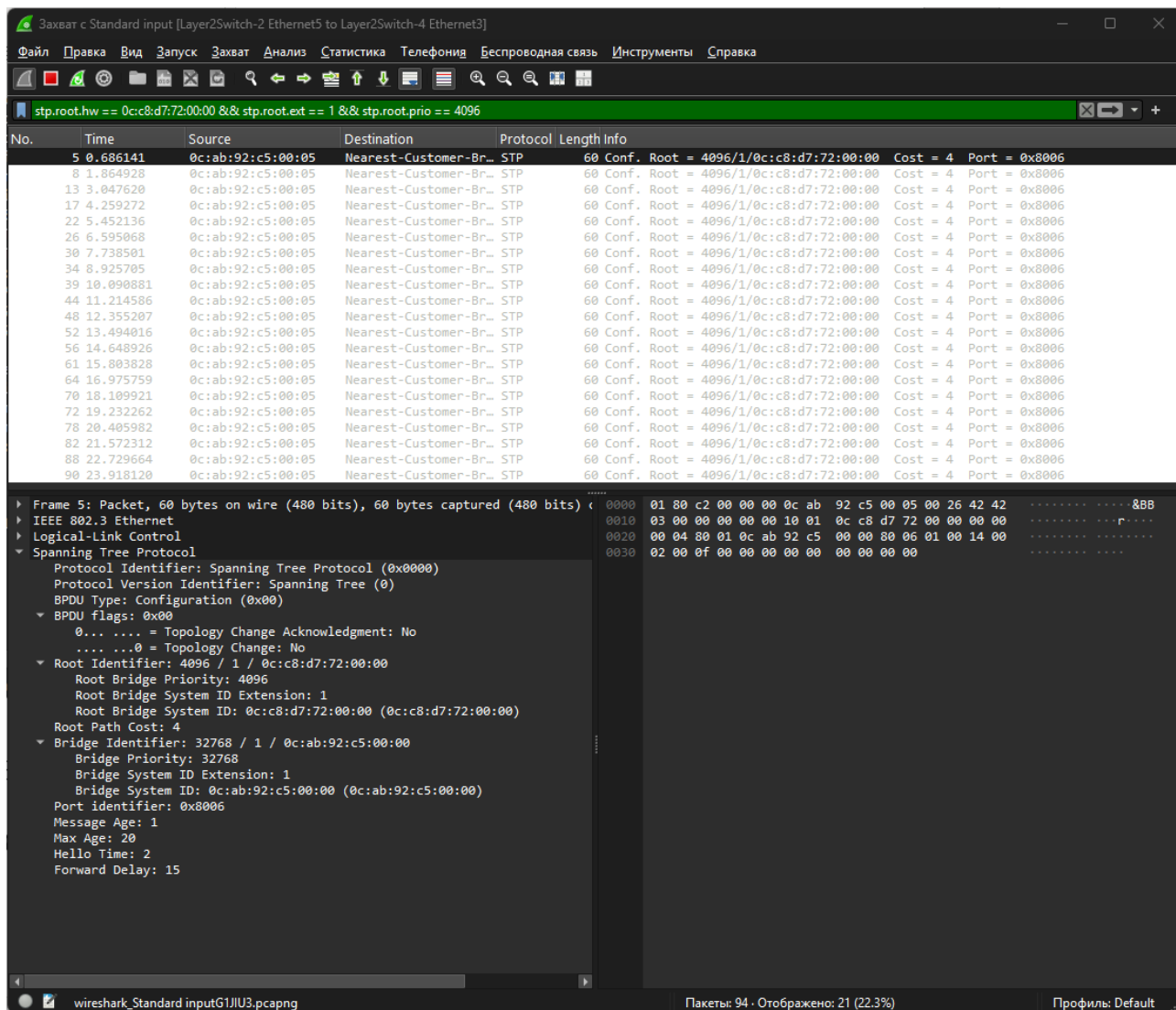


Рисунок 15 — STP Hello BPDU на линке Layer2Switch-2 Gi1/1 — Layer2Switch-4 Gi0/3

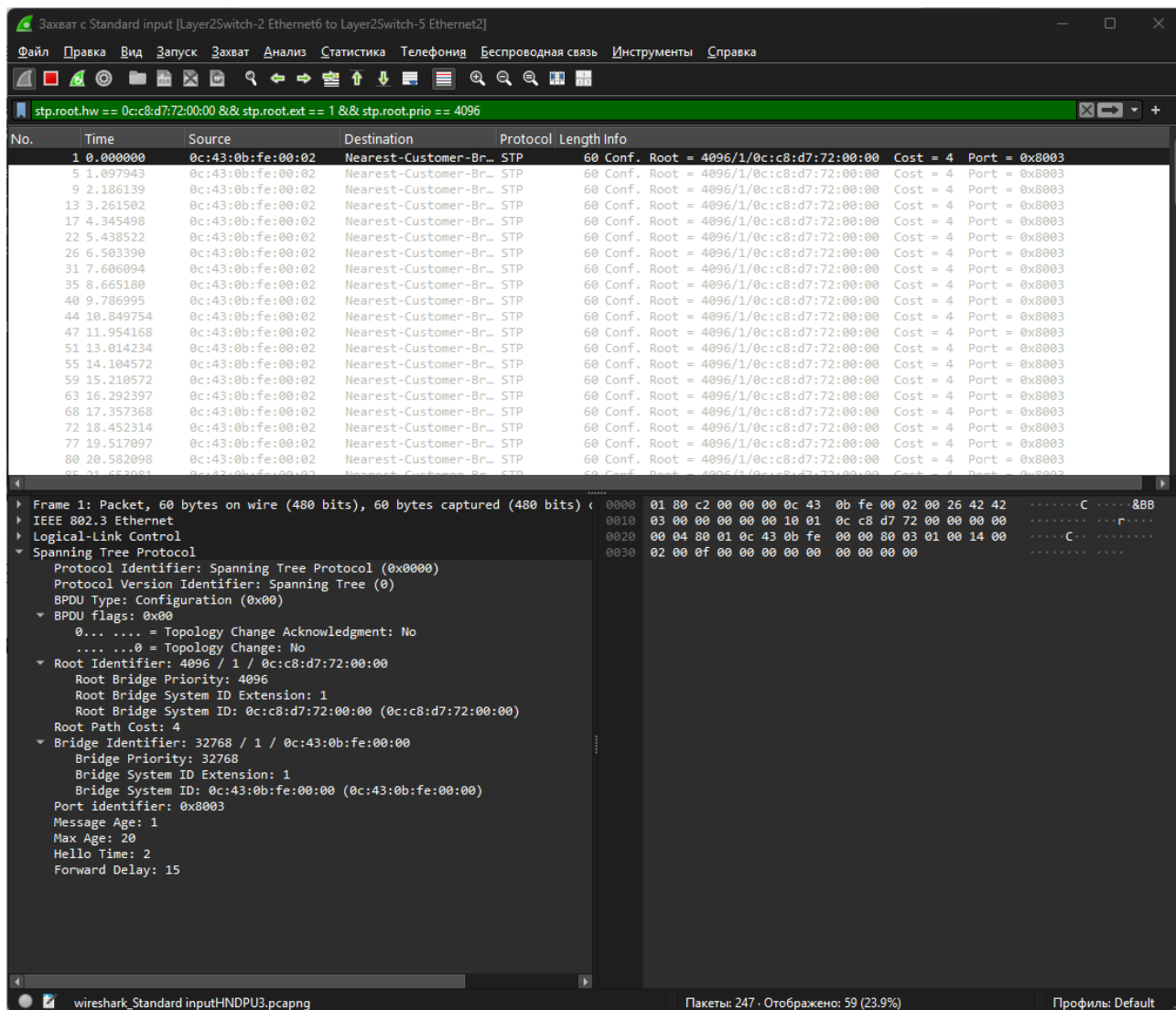


Рисунок 16 — STP Hello BPDUs на линке Layer2Switch-2 Gi1/2 — Layer2Switch-5 Gi0/2

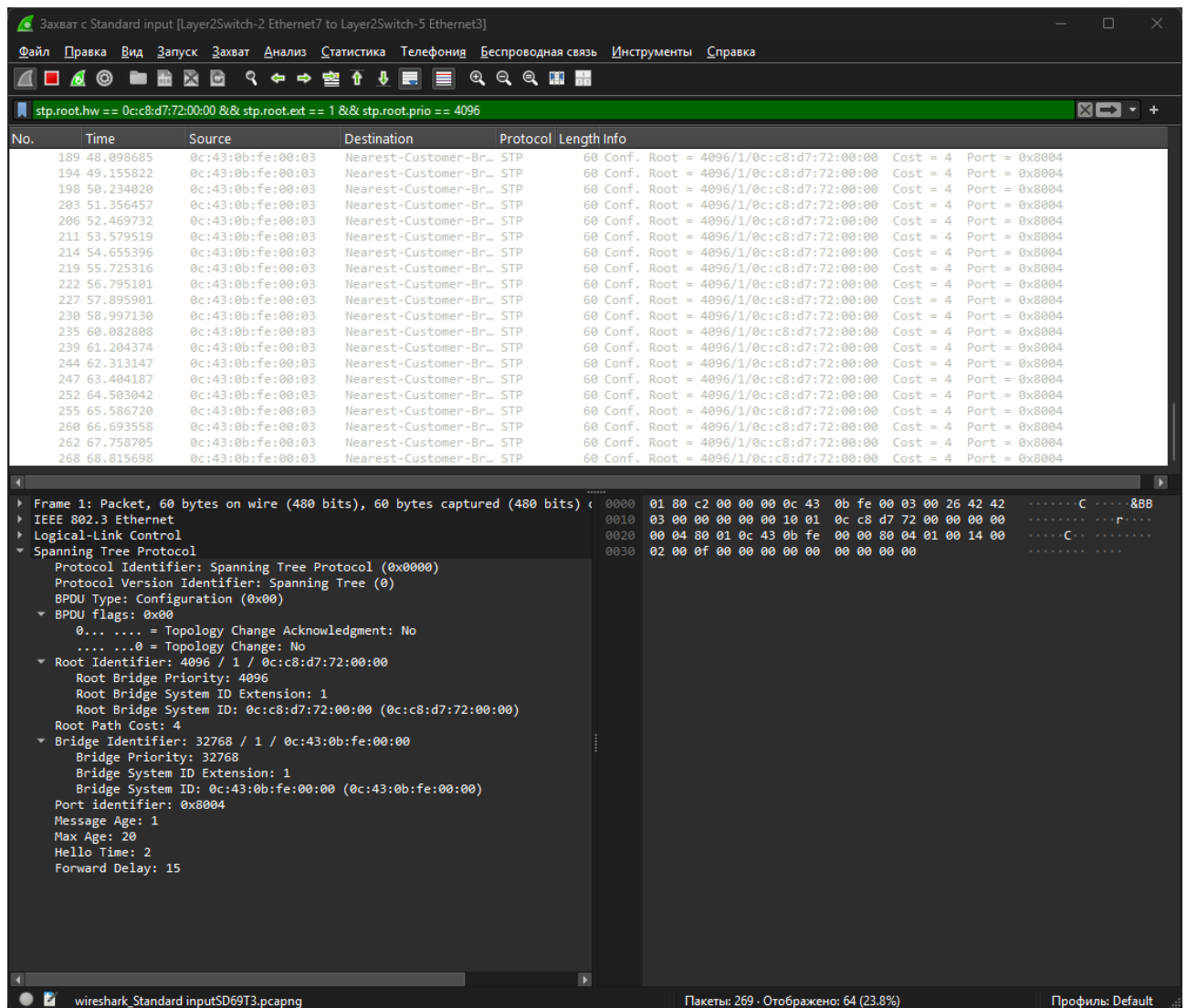


Рисунок 17 — STP Hello BPDU на линке Layer2Switch-2 Gi1/3 — Layer2Switch-5 Gi0/3

ЗАДАНИЕ 5

5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл

```
vIOS-L2-01>enable
```

```
vIOS-L2-01#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
vIOS-L2-01(config)#interface GigabitEthernet0/0
```

```
vIOS-L2-01(config-if)#spanning-tree vlan 1 cost 10
```

```
vIOS-L2-01(config-if)#end
```

```
vIOS-L2-01#
```

```
*Aug 17 13:00:48.390: %SYS-5-CONFIG_I: Configured from console by console
```

```
vIOS-L2-01#show spanning-tree vlan 1
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    4097
```

```
Address    0cc8.d772.0000
```

```
Cost        4
```

```
Port       2 (GigabitEthernet0/1)
```

```
Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
```

```
Address    0cd8.c8fe.0000
```

```
Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Aging Time  300 sec
```

```
Interface      Role Sts Cost    Prio.Nbr Type
```

```
-----
```

Gi0/0	Altn BLK 10	128.1	Shr
Gi0/1	Root FWD 4	128.2	Shr
Gi0/2	Altn BLK 4	128.3	Shr
Gi0/3	Altn BLK 4	128.4	Shr
Gi1/0	Desg FWD 4	128.5	Shr
Gi1/1	Desg FWD 4	128.6	Shr

vIOS-L2-01#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

Compressed configuration from 5072 bytes to 1941 bytes[OK]

*Aug 17 13:08:28.543: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...

*Aug 17 13:08:29.516: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

vIOS-L2-01>enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cc8.d772.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 4097 (priority 4096 sys-id-ext 1)

Address 0cc8.d772.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr
Gi2/0	Desg	FWD	4	128.9	Shr

vIOS-L2-01>enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cc8.d772.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0cab.92c5.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type

Gi0/0	Root	FWD	4	128.1		Shr
Gi0/1	Altn	BLK	4	128.2		Shr
Gi0/2	Altn	BLK	4	128.3		Shr
Gi0/3	Altn	BLK	4	128.4		Shr
Gi1/0	Desg	FWD	4	128.5		Shr
Gi1/1	Desg	FWD	4	128.6		Shr
Gi1/2	Altn	BLK	4	128.7		Shr
Gi1/3	Altn	BLK	4	128.8		Shr
Gi2/0	Desg	FWD	4	128.9		Shr

vIOS-L2-01>enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cc8.d772.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c34.c739.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type

Gi0/0	Root	FWD	4	128.1		Shr
Gi0/1	Altn	BLK	4	128.2		Shr
Gi0/2	Desg	FWD	4	128.3		Shr
Gi0/3	Desg	FWD	4	128.4		Shr
Gi1/0	Desg	FWD	4	128.5		Shr
Gi1/1	Desg	FWD	4	128.6		Shr

vIOS-L2-01>enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cc8.d772.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c43.0bfe.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type

Gi0/0	Root	FWD	4	128.1		Shr

Gi0/1	Altn BLK 4	128.2	Shr
Gi0/2	Desg FWD 4	128.3	Shr
Gi0/3	Desg FWD 4	128.4	Shr
Gi1/0	Desg FWD 4	128.5	Shr
Gi1/1	Desg FWD 4	128.6	Shr

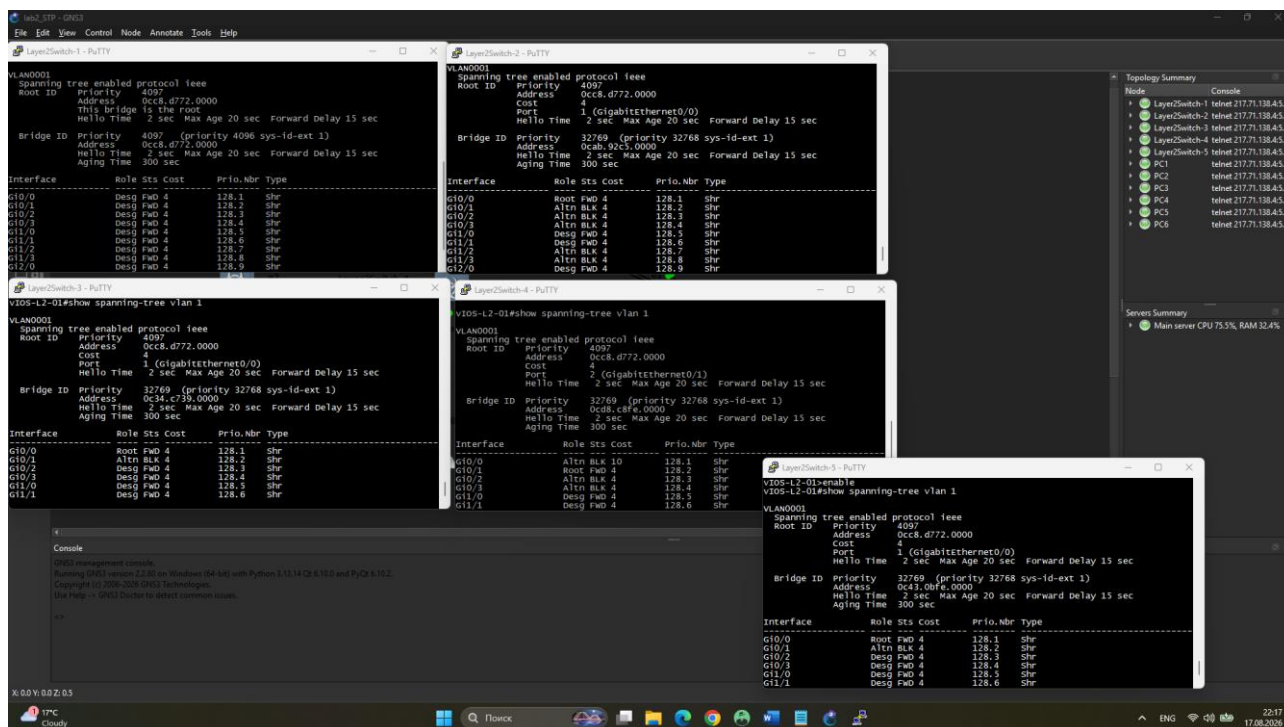


Рисунок 17.1 — Результаты проверки состояния STP на коммутаторах сети

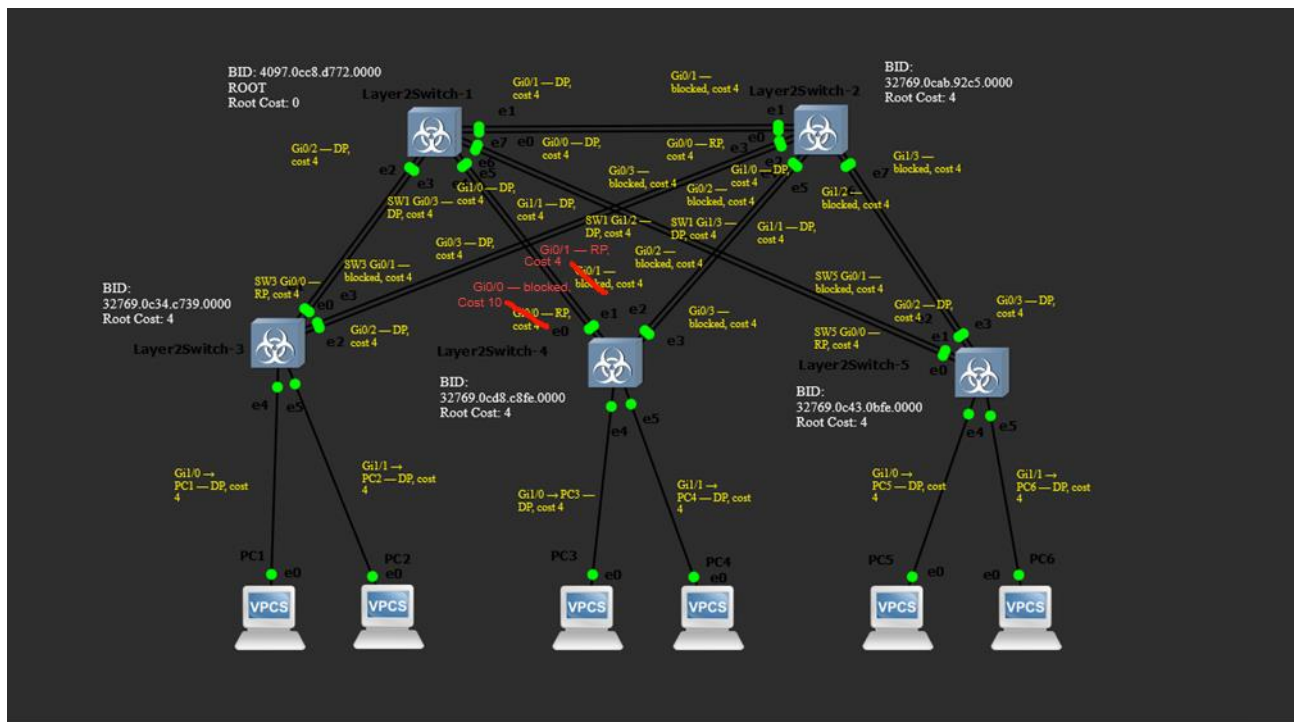


Рисунок 18 — Топология STP после изменения стоимости Root Port

ЗАДАНИЕ 6

б) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

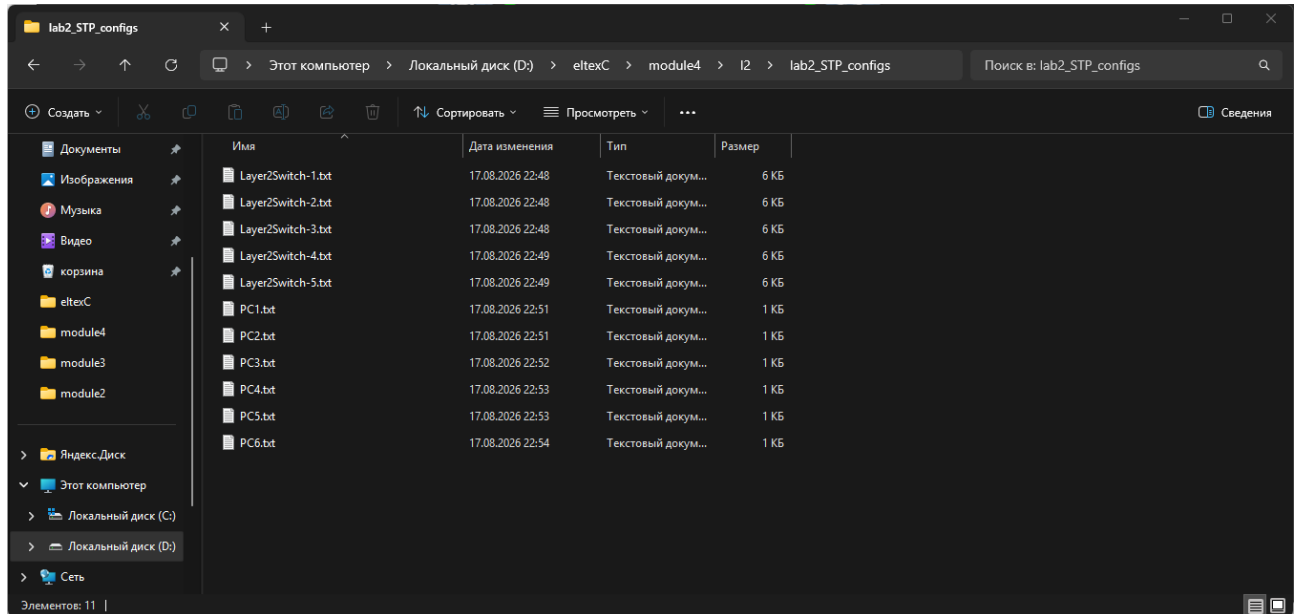


Рисунок 19 — Набор файлов конфигураций устройств

ЗАДАНИЕ 7

7*) Опциональное задание: заменить STP на RSTP (IEEE 802.1w), повторить 1-6, отметить резервные порты в п.3 и п.5, отличие работы протокола RSTP от протокола STP в п.4

```
vIOS-L2-01>enable
```

```
vIOS-L2-01#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
vIOS-L2-01(config)#spanning-tree mode rapid-pvst
```

```
vIOS-L2-01(config)#spanning-tree vlan 1 priority 4096
```

```
vIOS-L2-01(config)#end
```

```
vIOS-L2-01#
```

```
*Sep  7 13:12:06.469: %SYS-5-CONFIG_I: Configured from console by console
```

```
vIOS-L2-01#copy running-config startup-config
```

```
Destination filename [startup-config]?
```

```
Building configuration...
```

```
-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z  
8934B0z 837471z 867
```

```
-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z  
8934B0z 837471z 867
```

```
-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z  
8934B0z 837471z 867
```

```
-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z  
8934B0z 837471z 867
```

```
Compressed configuration from 5282 bytes to 2033 bytes
```

*Sep 7 13:12:19.164: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than.

*Sep 7 13:12:22.124: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than.

*Sep 7 13:12:24.696: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than.

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 889E8Dz 882

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 867

[OK]

*Sep 7 13:12:27.392: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than.

*Sep 7 13:12:30.308: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than.

*Sep 7 13:12:33.440: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than.

*Sep 7 13:12:36.820: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than.

*Sep 7 13:12:39.716: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than.

*Sep 7 13:12:40.674: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updat.

*Sep 7 13:12:45.607: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to.

vIOS-L2-01#

*Sep 7 13:12:58.378: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vios_l2-adn

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 4097

Address 0cc8.d772.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 4097 (priority 4096 sys-id-ext 1)

Address 0cc8.d772.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Desg	FWD	4	128.1	Shr	
Gi0/1	Desg	FWD	4	128.2	Shr	
Gi0/2	Desg	FWD	4	128.3	Shr	
Gi0/3	Desg	FWD	4	128.4	Shr	
Gi1/0	Desg	FWD	4	128.5	Shr	
Gi1/1	Desg	FWD	4	128.6	Shr	
Gi1/2	Desg	FWD	4	128.7	Shr	
Gi1/3	Desg	FWD	4	128.8	Shr	
Gi2/0	Desg	FWD	4	128.9	Shr	

vIOS-L2-01#

vIOS-L2-01>enable

vIOS-L2-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

vIOS-L2-01(config)#spanning-tree mode rapid-pvst

vIOS-L2-01(config)#end

vIOS-L2-01#

*Sep 7 13:15:05.934: %SYS-5-CONFIG_I: Configured from console by console

vIOS-L2-01#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 860

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 860

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

Compressed configuration from 5247 bytes to 2005 bytes

*Sep 7 13:15:11.527: %SYS-3-CPUHOG: Task is running for (2000)msecs,
more than.

*Sep 7 13:15:14.531: %SYS-3-CPUHOG: Task is running for (1999)msecs,
more than.

*Sep 7 13:15:17.419: %SYS-3-CPUHOG: Task is running for (1997)msecs,
more than.

*Sep 7 13:15:21.479: %SYS-3-CPUHOG: Task is running for (2000)msecs,
more than.

*Sep 7 13:15:25.127: %SYS-3-CPUHOG: Task is running for (1998)msecs,
more than.

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

[OK]

*Sep 7 13:15:28.783: %SYS-3-CPUHOG: Task is running for (1997)msecs,
more than.

*Sep 7 13:15:32.735: %SYS-3-CPUHOG: Task is running for (2000)msecs,
more than.

*Sep 7 13:15:35.090: %GRUB-5-CONFIG_WRITING: GRUB configuration
is being updat.

*Sep 7 13:15:37.053: %GRUB-5-CONFIG_WRITTEN: GRUB configuration
was written to.

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 32769

Address 0cab.92c5.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0cab.92c5.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type

Gi0/0	Desg	BLK	4	128.1	Shr
Gi0/1	Desg	BLK	4	128.2	Shr
Gi0/2	Desg	BLK	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	BLK	4	128.5	Shr
Gi1/1	Desg	BLK	4	128.6	Shr
Gi1/2	Desg	BLK	4	128.7	Shr
Gi1/3	Desg	BLK	4	128.8	Shr
Gi2/0	Desg	BLK	4	128.9	Shr

vIOS-L2-01#

vIOS-L2-01>enable

vIOS-L2-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

```

vIOS-L2-01(config)#spanning-tree mode rapid-pvst
vIOS-L2-01(config)#end
vIOS-L2-01#
*Sep  7 13:16:34.067: %SYS-5-CONFIG_I: Configured from console by
console
vIOS-L2-01#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 860
Compressed configuration from 5049 bytes to 1934 bytes
*Sep  7 13:16:40.308: %SYS-3-CPUHOG: Task is running for (1998)msecs,
more than.
*Sep  7 13:16:43.612: %SYS-3-CPUHOG: Task is running for (1998)msecs,
more than.
*Sep  7 13:16:47.388: %SYS-3-CPUHOG: Task is running for (1999)msecs,
more than.
-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

```

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

[OK]

*Sep 7 13:16:49.696: %SYS-3-CPUHOG: Task is running for (1999)msecs,
more than.

*Sep 7 13:16:53.336: %SYS-3-CPUHOG: Task is running for (2000)msecs,
more than.

*Sep 7 13:16:56.304: %SYS-3-CPUHOG: Task is running for (1999)msecs,
more than.

*Sep 7 13:16:58.812: %SYS-3-CPUHOG: Task is running for (1998)msecs,
more than.

*Sep 7 13:17:02.276: %SYS-3-CPUHOG: Task is running for (2000)msecs,
more than.

*Sep 7 13:17:03.662: %GRUB-5-CONFIG_WRITING: GRUB configuration
is being updat.

vIOS-L2-01#

*Sep 7 13:17:05.496: %GRUB-5-CONFIG_WRITTEN: GRUB configuration
was written to.

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 32769

Address 0c34.c739.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c34.c739.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Desg	BLK	4	128.1	Shr	
Gi0/1	Desg	FWD	4	128.2	Shr	
Gi0/2	Desg	BLK	4	128.3	Shr	
Gi0/3	Desg	BLK	4	128.4	Shr	
Gi1/0	Desg	BLK	4	128.5	Shr	
Gi1/1	Desg	BLK	4	128.6	Shr	

vIOS-L2-01#

vIOS-L2-01>enable

vIOS-L2-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

vIOS-L2-01(config)#interface GigabitEthernet0/0

vIOS-L2-01(config-if)#no spanning-tree vlan 1 cost 10

vIOS-L2-01(config-if)#end

vIOS-L2-01#

*Sep 7 13:06:34.427: %SYS-5-CONFIG_I: Configured from console by console

vIOS-L2-01#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async
write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async
write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async
write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async
write process", CPU hog, PC 0x00889D77

Compressed configuration from 5041 bytes to 1930 bytes

*Sep 7 13:08:05.134: %SYS-3-CPUHOG: Task is running for (2000)msecs,
more than (2000)msecs (167/150),process = Async write process.

*Sep 7 13:08:08.498: %SYS-3-CPUHOG: Task is running for (1998)msecs,
more than (2000)msecs (22/0),process = Async write process.

*Sep 7 13:08:10.646: %SYS-3-CPUHOG: Task is running for (2000)msecs,
more than (2000)msecs (38/24),process = Async write process.

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 86CE9Ez 8679B9z
8574D3z 85914Ez - Process "Exec", CPU hog, PC 0x00889D77

[OK]

*Sep 7 13:08:17.058: %SYS-3-CPUHOG: Task is running for (2000)msecs, more than (2000)msecs (32/24),process = Async write process.

*Sep 7 13:08:19.506: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (2000)msecs (13/1),process = Exec.

*Sep 7 13:08:22.551: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...

*Sep 7 13:08:26.759: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097

Address 0cc8.d772.0000

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0cd8.c8fe.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Root	FWD	4	128.1	Shr	
-------	------	-----	---	-------	-----	--

Gi0/1	Altn	BLK	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Altn	BLK	4	128.3	Shr	
-------	------	-----	---	-------	-----	--

Gi0/3	Altn	BLK	4	128.4	Shr	
-------	------	-----	---	-------	-----	--

Gi1/0	Desg	FWD	4	128.5	Shr	
-------	------	-----	---	-------	-----	--

Gi1/1 Desg FWD 4 128.6 Shr

vIOS-L2-01#

*Sep 7 13:11:36.068: %PLATFORM-5-SIGNATURE_VERIFIED: Image
'flash0:/vios_l2-adventerprisek9-m' passed code signing verification

vIOS-L2-01#enable

vIOS-L2-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

vIOS-L2-01(config)#spanning-tree mode rapid-pvst

vIOS-L2-01(config)#end

vIOS-L2-01#

*Sep 7 13:17:32.783: %SYS-5-CONFIG_I: Configured from console by
console

vIOS-L2-01#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async
write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async
write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async
write process", CPU hog, PC 0x008B5430

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x00889D77

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 86CE9Ez 8679B9z 8574D3z 85914Ez - Process "Exec", CPU hog, PC 0x008B5430

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 86B283z 86A3E7z 867C37z 86D0D4z 870997z - Process "Async write process", CPU hog, PC 0x008B5430

Compressed configuration from 5049 bytes to 1934 bytes[OK]

*Sep 7 13:17:38.310: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than (2000)msecs (28/17),process = Async write process.

*Sep 7 13:17:41.590: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than (2000)msecs (48/35),process = Async write process.

*Sep 7 13:17:44.818: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than (2000)msecs (14/0),process = Async write process.

*Sep 7 13:17:47.418: %SYS-3-CPUHOG: Task is running for (1999)msecs, more than (2000)msecs (9/0),process = Async write process.

*Sep 7 13:17:51.366: %SYS-3-CPUHOG: Task is running for (2000)msecs, more than (2000)msecs (30/20),process = Async write process.

*Sep 7 13:17:54.594: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than (2000)msecs (7/0),process = Async write process.

*Sep 7 13:17:57.398: %SYS-3-CPUHOG: Task is running for (1997)msecs, more than (2000)msecs (16/0),process = Exec.

*Sep 7 13:18:00.802: %SYS-3-CPUHOG: Task is running for (1998)msecs, more than (2000)msecs (34/25),process = Async write process.

*Sep 7 13:18:02.397: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...

vIOS-L2-01#

*Sep 7 13:18:03.952: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 32769

Address 0cab.92c5.0000

Cost 4

Port 3 (GigabitEthernet0/2)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0cd8.c8fe.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Desg	BLK	4	128.1	Shr	
-------	------	-----	---	-------	-----	--

Gi0/1	Desg	BLK	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Root FWD 4	128.3	Shr
Gi0/3	Altn BLK 4	128.4	Shr
Gi1/0	Desg BLK 4	128.5	Shr
Gi1/1	Desg BLK 4	128.6	Shr

vIOS-L2-01>enable

vIOS-L2-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

vIOS-L2-01(config)#spanning-tree mode rapid-pvst

vIOS-L2-01(config)#end

vIOS-L2-01#

*Sep 7 13:18:08.262: %SYS-5-CONFIG_I: Configured from console by console

vIOS-L2-01#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 860

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z 8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

Compressed configuration from 5049 bytes to 1934 bytes

*Sep 7 13:18:13.859: %SYS-3-CPUHOG: Task is running for (1999)msecs,
more than.

*Sep 7 13:18:17.215: %SYS-3-CPUHOG: Task is running for (1998)msecs,
more than.

*Sep 7 13:18:20.231: %SYS-3-CPUHOG: Task is running for (1999)msecs,
more than.

*Sep 7 13:18:24.183: %SYS-3-CPUHOG: Task is running for (1997)msecs,
more than.

*Sep 7 13:18:26.211: %SYS-3-CPUHOG: Task is running for (3997)msecs,
more than.

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

-Traceback= 1C489D8z 88DD15z 8B5BAEz 8B58E0z 8B554Dz 8958B2z
8934B0z 837471z 867

[OK]

*Sep 7 13:18:28.807: %SYS-3-CPUHOG: Task is running for (1998)msecs,
more than.

*Sep 7 13:18:31.435: %SYS-3-CPUHOG: Task is running for (2000)msecs,
more than.

*Sep 7 13:18:34.667: %SYS-3-CPUHOG: Task is running for (1998)msecs,
more than.

*Sep 7 13:18:38.469: %GRUB-5-CONFIG_WRITING: GRUB configuration
is being updat.

*Sep 7 13:18:40.210: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to.

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 32769

Address 0c34.c739.0000

Cost 8

Port 3 (GigabitEthernet0/2)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c43.0bfe.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Desg	BLK	4	128.1	Shr	
-------	------	-----	---	-------	-----	--

Gi0/1	Desg	BLK	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Root	FWD	4	128.3	Shr	
-------	------	-----	---	-------	-----	--

Gi0/3	Altn	BLK	4	128.4	Shr	
-------	------	-----	---	-------	-----	--

Gi1/0	Desg	BLK	4	128.5	Shr	
-------	------	-----	---	-------	-----	--

Gi1/1	Desg	BLK	4	128.6	Shr	
-------	------	-----	---	-------	-----	--

на каждом из пяти коммутаторов заново выполни только:

enable

show spanning-tree vlan 1

vIOS-L2-01#enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 4097

Address 0cc8.d772.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 4097 (priority 4096 sys-id-ext 1)

Address 0cc8.d772.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi0/0	Desg	FWD	4	128.1	Shr
-------	------	-----	---	-------	-----

Gi0/1	Desg	FWD	4	128.2	Shr
-------	------	-----	---	-------	-----

Gi0/2	Desg	FWD	4	128.3	Shr
-------	------	-----	---	-------	-----

Gi0/3	Desg	FWD	4	128.4	Shr
-------	------	-----	---	-------	-----

Gi1/0	Desg	FWD	4	128.5	Shr
-------	------	-----	---	-------	-----

Gi1/1	Desg	FWD	4	128.6	Shr
-------	------	-----	---	-------	-----

Gi1/2	Desg	FWD	4	128.7	Shr
-------	------	-----	---	-------	-----

Gi1/3	Desg	FWD	4	128.8	Shr
-------	------	-----	---	-------	-----

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi2/0	Desg	FWD	4	128.9	Shr
-------	------	-----	---	-------	-----

vIOS-L2-01#enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 32769

 Address 0c43.0bfe.0000

 Cost 4

 Port 7 (GigabitEthernet1/2)

 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

 Address 0cab.92c5.0000

 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Desg	BLK	4	128.1	Shr	
-------	------	-----	---	-------	-----	--

Gi0/1	Desg	BLK	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Desg	BLK	4	128.3	Shr	
-------	------	-----	---	-------	-----	--

Gi0/3	Desg	BLK	4	128.4	Shr	
-------	------	-----	---	-------	-----	--

Gi1/0	Desg	BLK	4	128.5	Shr	
-------	------	-----	---	-------	-----	--

Gi1/1	Desg	BLK	4	128.6	Shr	
-------	------	-----	---	-------	-----	--

Gi1/2	Root	FWD	4	128.7	Shr	
-------	------	-----	---	-------	-----	--

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi1/3	Altn BLK 4	128.8	Shr
Gi2/0	Desg BLK 4	128.9	Shr

vIOS-L2-01# enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 32769

Address 0c34.c739.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c34.c739.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role Sts Cost	Prio.Nbr	Type
-----------	---------------	----------	------

Gi0/0	Desg FWD 4	128.1	Shr
Gi0/1	Desg BLK 4	128.2	Shr
Gi0/2	Desg BLK 4	128.3	Shr
Gi0/3	Desg BLK 4	128.4	Shr
Gi1/0	Desg BLK 4	128.5	Shr
Gi1/1	Desg BLK 4	128.6	Shr

vIOS-L2-01#enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 4097

 Address 0cc8.d772.0000

 Cost 4

 Port 1 (GigabitEthernet0/0)

 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

 Address 0cd8.c8fe.0000

 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Root	FWD	4	128.1	Shr	
-------	------	-----	---	-------	-----	--

Gi0/1	Altn	BLK	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Desg	BLK	4	128.3	Shr	
-------	------	-----	---	-------	-----	--

Gi0/3	Desg	BLK	4	128.4	Shr	
-------	------	-----	---	-------	-----	--

Gi1/0	Desg	BLK	4	128.5	Shr	
-------	------	-----	---	-------	-----	--

Gi1/1	Desg	BLK	4	128.6	Shr	
-------	------	-----	---	-------	-----	--

vIOS-L2-01#enable

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 32769

Address 0c43.0bfe.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c43.0bfe.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Desg	BLK	4	128.1	Shr	
-------	------	-----	---	-------	-----	--

Gi0/1	Desg	BLK	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Desg	FWD	4	128.3	Shr	
-------	------	-----	---	-------	-----	--

Gi0/3	Desg	BLK	4	128.4	Shr	
-------	------	-----	---	-------	-----	--

Gi1/0	Desg	BLK	4	128.5	Shr	
-------	------	-----	---	-------	-----	--

Gi1/1	Desg	BLK	4	128.6	Shr	
-------	------	-----	---	-------	-----	--

vIOS-L2-01#enable

vIOS-L2-01#show spanning-tree inconsistentports

Name	Interface	Inconsistency
------	-----------	---------------

Number of inconsistent ports (segments) in the system : 0

vIOS-L2-01#show spanning-tree vlan 1 detail

VLAN0001 is executing the rstp compatible Spanning Tree protocol
Bridge Identifier has priority 32768, sysid 1, address 0c34.c739.0000
Configured hello time 2, max age 20, forward delay 15, transmit hold-count 6
We are the root of the spanning tree
Topology change flag set, detected flag not set
Number of topology changes 1167 last change occurred 00:00:02 ago
from GigabitEthernet0/0
Times: hold 1, topology change 35, notification 2
hello 2, max age 20, forward delay 15
Timers: hello 0, topology change 32, notification 0, aging 300

Port 1 (GigabitEthernet0/0) of VLAN0001 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.1.
Designated root has priority 32769, address 0c34.c739.0000
Designated bridge has priority 32769, address 0c34.c739.0000
Designated port id is 128.1, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 392
Link type is shared by default
BPDU: sent 1962, received 1756

Port 2 (GigabitEthernet0/1) of VLAN0001 is designated blocking
Port path cost 4, Port priority 128, Port Identifier 128.2.
Designated root has priority 32769, address 0c34.c739.0000
Designated bridge has priority 32769, address 0c34.c739.0000
Designated port id is 128.2, designated path cost 0
Timers: message age 0, forward delay 13, hold 0

Number of transitions to forwarding state: 363

Link type is shared by default

BPDU: sent 1886, received 1756

Port 3 (GigabitEthernet0/2) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.3.

Designated root has priority 32769, address 0c34.c739.0000

Designated bridge has priority 32769, address 0c34.c739.0000

Designated port id is 128.3, designated path cost 0

Timers: message age 0, forward delay 10, hold 0

Number of transitions to forwarding state: 175

Link type is shared by default

BPDU: sent 1944, received 4636

Port 4 (GigabitEthernet0/3) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.4.

Designated root has priority 32769, address 0c34.c739.0000

Designated bridge has priority 32769, address 0c34.c739.0000

Designated port id is 128.4, designated path cost 0

Timers: message age 0, forward delay 7, hold 0

Number of transitions to forwarding state: 141

Link type is shared by default

BPDU: sent 1856, received 4302

Port 5 (GigabitEthernet1/0) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.5.

Designated root has priority 32769, address 0c34.c739.0000

Designated bridge has priority 32769, address 0c34.c739.0000

Designated port id is 128.5, designated path cost 0

Timers: message age 0, forward delay 3, hold 0

Number of transitions to forwarding state: 0

Link type is shared by default

BPDU: sent 1963, received 0

Port 6 (GigabitEthernet1/1) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.6.

Designated root has priority 32769, address 0c34.c739.0000

Designated bridge has priority 32769, address 0c34.c739.0000

Designated port id is 128.6, designated path cost 0

Timers: message age 0, forward delay 3, hold 0

Number of transitions to forwarding state: 0

Link type is shared by default

BPDU: sent 1963, received 0

vIOS-L2-01#enable

vIOS-L2-01#show spanning-tree inconsistentports

Name	Interface	Inconsistency
------	-----------	---------------

Number of inconsistent ports (segments) in the system : 0

vIOS-L2-01#show spanning-tree vlan 1 detail

VLAN0001 is executing the rstp compatible Spanning Tree protocol

Bridge Identifier has priority 32768, sysid 1, address 0c43.0bfe.0000

Configured hello time 2, max age 20, forward delay 15, transmit hold-count 6

We are the root of the spanning tree

Topology change flag set, detected flag not set

Number of topology changes 1718 last change occurred 00:00:00
from GigabitEthernet0/3

Times: hold 1, topology change 35, notification 2
hello 2, max age 20, forward delay 15

Timers: hello 0, topology change 34, notification 0, aging 300

Port 1 (GigabitEthernet0/0) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.1.

Designated root has priority 32769, address 0c43.0bfe.0000

Designated bridge has priority 32769, address 0c43.0bfe.0000

Designated port id is 128.1, designated path cost 0

Timers: message age 0, forward delay 13, hold 0

Number of transitions to forwarding state: 388

Link type is shared by default

BPDU: sent 2559, received 1646

Port 2 (GigabitEthernet0/1) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.2.

Designated root has priority 32769, address 0c43.0bfe.0000

Designated bridge has priority 32769, address 0c43.0bfe.0000

Designated port id is 128.2, designated path cost 0

Timers: message age 0, forward delay 13, hold 0

Number of transitions to forwarding state: 330

Link type is shared by default

BPDU: sent 2446, received 1646

Port 3 (GigabitEthernet0/2) of VLAN0001 is root forwarding

Port path cost 4, Port priority 128, Port Identifier 128.3.

Designated root has priority 32769, address 0c34.c739.0000

Designated bridge has priority 32769, address 0cab.92c5.0000

Designated port id is 128.7, designated path cost 4
Timers: message age 15, forward delay 0, hold 0
Number of transitions to forwarding state: 500
Link type is shared by default
BPDU: sent 2515, received 4271

Port 4 (GigabitEthernet0/3) of VLAN0001 is alternate blocking
Port path cost 4, Port priority 128, Port Identifier 128.4.
Designated root has priority 32769, address 0c34.c739.0000
Designated bridge has priority 32769, address 0cab.92c5.0000
Designated port id is 128.8, designated path cost 4
Timers: message age 15, forward delay 0, hold 0
Number of transitions to forwarding state: 415
Link type is shared by default
BPDU: sent 2224, received 4008

Port 5 (GigabitEthernet1/0) of VLAN0001 is designated blocking
Port path cost 4, Port priority 128, Port Identifier 128.5.
Designated root has priority 32769, address 0c43.0bfe.0000
Designated bridge has priority 32769, address 0c43.0bfe.0000
Designated port id is 128.5, designated path cost 0
Timers: message age 0, forward delay 5, hold 0
Number of transitions to forwarding state: 0
Link type is shared by default
BPDU: sent 2566, received 0

Port 6 (GigabitEthernet1/1) of VLAN0001 is designated blocking
Port path cost 4, Port priority 128, Port Identifier 128.6.
Designated root has priority 32769, address 0c43.0bfe.0000
Designated bridge has priority 32769, address 0c43.0bfe.0000

Designated port id is 128.6, designated path cost 0
Timers: message age 0, forward delay 5, hold 0
Number of transitions to forwarding state: 0
Link type is shared by default
BPDU: sent 2566, received 0

vIOS-L2-01#show spanning-tree interface GigabitEthernet0/2 detail

Port 3 (GigabitEthernet0/2) of VLAN0001 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.3.

Designated root has priority 4097, address 0cc8.d772.0000

Designated bridge has priority 4097, address 0cc8.d772.0000

Designated port id is 128.3, designated path cost 0, Topology change is set

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 4

Link type is shared by default

BPDU: sent 600, received 4201

Port 3 (GigabitEthernet0/2) of VLAN0100 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.3.

Designated root has priority 32868, address 0cc8.d772.0000

Designated bridge has priority 32868, address 0cc8.d772.0000

Designated port id is 128.3, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

Link type is shared by default

BPDU: sent 573, received 0

Port 3 (GigabitEthernet0/2) of VLAN0200 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.3.

Designated root has priority 32968, address 0cc8.d772.0000
Designated bridge has priority 32968, address 0cc8.d772.0000
Designated port id is 128.3, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is shared by default
BPDU: sent 573, received 0

Port 3 (GigabitEthernet0/2) of VLAN0300 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.3.
Designated root has priority 33068, address 0cc8.d772.0000
Designated bridge has priority 33068, address 0cc8.d772.0000
Designated port id is 128.3, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is shared by default
BPDU: sent 573, received 0

vIOS-L2-01# show spanning-tree interface GigabitEthernet0/3 detail

Port 4 (GigabitEthernet0/3) of VLAN0001 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.4.
Designated root has priority 4097, address 0cc8.d772.0000
Designated bridge has priority 4097, address 0cc8.d772.0000
Designated port id is 128.4, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 4
Link type is shared by default
BPDU: sent 603, received 4097

Port 4 (GigabitEthernet0/3) of VLAN0100 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.4.

Designated root has priority 32868, address 0cc8.d772.0000
Designated bridge has priority 32868, address 0cc8.d772.0000
Designated port id is 128.4, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is shared by default
BPDU: sent 576, received 0

Port 4 (GigabitEthernet0/3) of VLAN0200 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.4.
Designated root has priority 32968, address 0cc8.d772.0000
Designated bridge has priority 32968, address 0cc8.d772.0000
Designated port id is 128.4, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is shared by default
BPDU: sent 576, received 0

Port 4 (GigabitEthernet0/3) of VLAN0300 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.4.
Designated root has priority 33068, address 0cc8.d772.0000
Designated bridge has priority 33068, address 0cc8.d772.0000
Designated port id is 128.4, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is shared by default
BPDU: sent 575, received 0

vIOS-L2-01#show spanning-tree interface GigabitEthernet1/2 detail
Port 7 (GigabitEthernet1/2) of VLAN0001 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.7.

Designated root has priority 4097, address 0cc8.d772.0000
Designated bridge has priority 4097, address 0cc8.d772.0000
Designated port id is 128.7, designated path cost 0, Topology change is set
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 4
Link type is shared by default
BPDU: sent 605, received 4847

Port 7 (GigabitEthernet1/2) of VLAN0100 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.7.
Designated root has priority 32868, address 0cc8.d772.0000
Designated bridge has priority 32868, address 0cc8.d772.0000
Designated port id is 128.7, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is shared by default
BPDU: sent 578, received 0

Port 7 (GigabitEthernet1/2) of VLAN0200 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.7.
Designated root has priority 32968, address 0cc8.d772.0000
Designated bridge has priority 32968, address 0cc8.d772.0000
Designated port id is 128.7, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is shared by default
BPDU: sent 578, received 0

Port 7 (GigabitEthernet1/2) of VLAN0300 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.7.

Designated root has priority 33068, address 0cc8.d772.0000
Designated bridge has priority 33068, address 0cc8.d772.0000
Designated port id is 128.7, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is shared by default
BPDU: sent 578, received 0

vIOS-L2-01#show spanning-tree interface GigabitEthernet1/3 detail

Port 8 (GigabitEthernet1/3) of VLAN0001 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.8.

Designated root has priority 4097, address 0cc8.d772.0000

Designated bridge has priority 4097, address 0cc8.d772.0000

Designated port id is 128.8, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 4

Link type is shared by default

BPDU: sent 606, received 4677

Port 8 (GigabitEthernet1/3) of VLAN0100 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.8.

Designated root has priority 32868, address 0cc8.d772.0000

Designated bridge has priority 32868, address 0cc8.d772.0000

Designated port id is 128.8, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

Link type is shared by default

BPDU: sent 579, received 0

Port 8 (GigabitEthernet1/3) of VLAN0200 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.8.

Designated root has priority 32968, address 0cc8.d772.0000
Designated bridge has priority 32968, address 0cc8.d772.0000
Designated port id is 128.8, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is shared by default
BPDU: sent 579, received 0

Port 8 (GigabitEthernet1/3) of VLAN0300 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.8.
Designated root has priority 33068, address 0cc8.d772.0000
Designated bridge has priority 33068, address 0cc8.d772.0000
Designated port id is 128.8, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is shared by default
BPDU: sent 579, received 0
vIOS-L2-01#

vIOS-L2-01#show spanning-tree vlan 1 detail

VLAN0001 is executing the rstp compatible Spanning Tree protocol
Bridge Identifier has priority 32768, sysid 1, address 0c34.c739.0000
Configured hello time 2, max age 20, forward delay 15, transmit hold-count 6
We are the root of the spanning tree
Topology change flag set, detected flag not set
Number of topology changes 1167 last change occurred 00:00:02 ago
from GigabitEthernet0/0
Times: hold 1, topology change 35, notification 2

hello 2, max age 20, forward delay 15

Timers: hello 0, topology change 32, notification 0, aging 300

Port 1 (GigabitEthernet0/0) of VLAN0001 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.1.

Designated root has priority 32769, address 0c34.c739.0000

Designated bridge has priority 32769, address 0c34.c739.0000

Designated port id is 128.1, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 392

Link type is shared by default

BPDU: sent 1962, received 1756

Port 2 (GigabitEthernet0/1) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.2.

Designated root has priority 32769, address 0c34.c739.0000

Designated bridge has priority 32769, address 0c34.c739.0000

Designated port id is 128.2, designated path cost 0

Timers: message age 0, forward delay 13, hold 0

Number of transitions to forwarding state: 363

Link type is shared by default

BPDU: sent 1886, received 1756

Port 3 (GigabitEthernet0/2) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.3.

Designated root has priority 32769, address 0c34.c739.0000

Designated bridge has priority 32769, address 0c34.c739.0000

Designated port id is 128.3, designated path cost 0

Timers: message age 0, forward delay 10, hold 0

Number of transitions to forwarding state: 175

Link type is shared by default

BPDU: sent 1944, received 4636

Port 4 (GigabitEthernet0/3) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.4.

Designated root has priority 32769, address 0c34.c739.0000

Designated bridge has priority 32769, address 0c34.c739.0000

Designated port id is 128.4, designated path cost 0

Timers: message age 0, forward delay 7, hold 0

Number of transitions to forwarding state: 141

Link type is shared by default

BPDU: sent 1856, received 4302

Port 5 (GigabitEthernet1/0) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.5.

Designated root has priority 32769, address 0c34.c739.0000

Designated bridge has priority 32769, address 0c34.c739.0000

Designated port id is 128.5, designated path cost 0

Timers: message age 0, forward delay 3, hold 0

Number of transitions to forwarding state: 0

Link type is shared by default

BPDU: sent 1963, received 0

Port 6 (GigabitEthernet1/1) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.6.

Designated root has priority 32769, address 0c34.c739.0000

Designated bridge has priority 32769, address 0c34.c739.0000

Designated port id is 128.6, designated path cost 0

Timers: message age 0, forward delay 3, hold 0

Number of transitions to forwarding state: 0

Link type is shared by default

BPDU: sent 1963, received 0

vIOS-L2-01#show spanning-tree interface GigabitEthernet0/0 detail

Port 1 (GigabitEthernet0/0) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.1.

Designated root has priority 32769, address 0c34.c739.0000

Designated bridge has priority 32769, address 0c34.c739.0000

Designated port id is 128.1, designated path cost 0

Timers: message age 0, forward delay 12, hold 0

Number of transitions to forwarding state: 503

Link type is shared by default

BPDU: sent 2485, received 2239

Port 1 (GigabitEthernet0/0) of VLAN0100 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.1.

Designated root has priority 32868, address 0c34.c739.0000

Designated bridge has priority 32868, address 0c34.c739.0000

Designated port id is 128.1, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

Link type is shared by default

BPDU: sent 700, received 0

Port 1 (GigabitEthernet0/0) of VLAN0200 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.1.

Designated root has priority 32968, address 0c34.c739.0000

Designated bridge has priority 32968, address 0c34.c739.0000

Designated port id is 128.1, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

Link type is shared by default

BPDU: sent 700, received 0

Port 1 (GigabitEthernet0/0) of VLAN0300 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.1.

Designated root has priority 33068, address 0c34.c739.0000

Designated bridge has priority 33068, address 0c34.c739.0000

Designated port id is 128.1, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

Link type is shared by default

BPDU: sent 700, received 0

vIOS-L2-01#show spanning-tree interface GigabitEthernet0/1 detail

Port 2 (GigabitEthernet0/1) of VLAN0001 is alternate blocking

Port path cost 4, Port priority 128, Port Identifier 128.2.

Designated root has priority 4097, address 0cc8.d772.0000

Designated bridge has priority 4097, address 0cc8.d772.0000

Designated port id is 128.4, designated path cost 0

Timers: message age 15, forward delay 0, hold 0

Number of transitions to forwarding state: 466

Link type is shared by default

BPDU: sent 2399, received 2252

Port 2 (GigabitEthernet0/1) of VLAN0100 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.2.

Designated root has priority 32868, address 0c34.c739.0000

Designated bridge has priority 32868, address 0c34.c739.0000

Designated port id is 128.2, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

Link type is shared by default

BPDU: sent 703, received 0

Port 2 (GigabitEthernet0/1) of VLAN0200 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.2.

Designated root has priority 32968, address 0c34.c739.0000

Designated bridge has priority 32968, address 0c34.c739.0000

Designated port id is 128.2, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

Link type is shared by default

BPDU: sent 703, received 0

Port 2 (GigabitEthernet0/1) of VLAN0300 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.2.

Designated root has priority 33068, address 0c34.c739.0000

Designated bridge has priority 33068, address 0c34.c739.0000

Designated port id is 128.2, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

Link type is shared by default

BPDU: sent 703, received 0

vIOS-L2-01#

vIOS-L2-01#show spanning-tree interface GigabitEthernet0/0 detail

Port 1 (GigabitEthernet0/0) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.1.

Designated root has priority 32769, address 0c43.0bfe.0000

Designated bridge has priority 32769, address 0c43.0bfe.0000
Designated port id is 128.1, designated path cost 0
Timers: message age 0, forward delay 12, hold 0
Number of transitions to forwarding state: 509
Link type is shared by default
BPDU: sent 3341, received 2149

Port 1 (GigabitEthernet0/0) of VLAN0100 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.1.
Designated root has priority 32868, address 0c43.0bfe.0000
Designated bridge has priority 32868, address 0c43.0bfe.0000
Designated port id is 128.1, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is shared by default
BPDU: sent 667, received 0

Port 1 (GigabitEthernet0/0) of VLAN0200 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.1.
Designated root has priority 32968, address 0c43.0bfe.0000
Designated bridge has priority 32968, address 0c43.0bfe.0000
Designated port id is 128.1, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
Link type is shared by default
BPDU: sent 667, received 0

Port 1 (GigabitEthernet0/0) of VLAN0300 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.1.
Designated root has priority 33068, address 0c43.0bfe.0000

Designated bridge has priority 33068, address 0c43.0bfe.0000

Designated port id is 128.1, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

Link type is shared by default

BPDUs: sent 667, received 0

vIOS-L2-01#show spanning-tree interface GigabitEthernet0/1 detail

Port 2 (GigabitEthernet0/1) of VLAN0001 is designated blocking

Port path cost 4, Port priority 128, Port Identifier 128.2.

Designated root has priority 32769, address 0c43.0bfe.0000

Designated bridge has priority 32769, address 0c43.0bfe.0000

Designated port id is 128.2, designated path cost 0

Timers: message age 0, forward delay 13, hold 0

Number of transitions to forwarding state: 433

Link type is shared by default

BPDUs: sent 3206, received 2157

Port 2 (GigabitEthernet0/1) of VLAN0100 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.2.

Designated root has priority 32868, address 0c43.0bfe.0000

Designated bridge has priority 32868, address 0c43.0bfe.0000

Designated port id is 128.2, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

Link type is shared by default

BPDUs: sent 669, received 0

Port 2 (GigabitEthernet0/1) of VLAN0200 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.2.

Designated root has priority 32968, address 0c43.0bfe.0000

Designated bridge has priority 32968, address 0c43.0bfe.0000

Designated port id is 128.2, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

Link type is shared by default

BPDU: sent 669, received 0

Port 2 (GigabitEthernet0/1) of VLAN0300 is designated forwarding

Port path cost 4, Port priority 128, Port Identifier 128.2.

Designated root has priority 33068, address 0c43.0bfe.0000

Designated bridge has priority 33068, address 0c43.0bfe.0000

Designated port id is 128.2, designated path cost 0

Timers: message age 0, forward delay 0, hold 0

Number of transitions to forwarding state: 1

Link type is shared by default

BPDU: sent 669, received 0

vIOS-L2-01#

```
vIOS-L2-01>enable
```

```
vIOS-L2-01#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
vIOS-L2-01(config)#spanning-tree mode rapid-pvst
```

```
vIOS-L2-01(config)#spanning-tree vlan 1 priority 4096
```

```
vIOS-L2-01(config)#end
```

```
vIOS-L2-01#
```

*Sep 7 13:45:10.719: %SYS-5-CONFIG_I: Configured from console by console

```
vIOS-L2-01#show spanning-tree vlan 1
```

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 4097

Address 0cc8.d772.0000

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 4097 (priority 4096 sys-id-ext 1)

Address 0cc8.d772.0000

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi0/0	Desg	FWD	4	128.1	Shr	
-------	------	-----	---	-------	-----	--

Gi0/1	Desg	FWD	4	128.2	Shr	
-------	------	-----	---	-------	-----	--

Gi0/2	Desg	FWD	4	128.3	Shr	
-------	------	-----	---	-------	-----	--

Gi0/3	Desg	FWD	4	128.4	Shr	
-------	------	-----	---	-------	-----	--

Gi1/0	Desg	FWD	4	128.5	Shr	
-------	------	-----	---	-------	-----	--

Gi1/1	Desg FWD 4	128.6	Shr
Gi1/2	Desg FWD 4	128.7	Shr
Gi1/3	Desg FWD 4	128.8	Shr

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

Gi2/0	Desg FWD 4	128.9	Shr
-------	------------	-------	-----
